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**Neurobehavioral effects of low to moderate organophosphate
exposure in lawn and landscape applicators**

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ABSTRACT

**Title of Dissertation: NEUROBEHAVIORAL EFFECTS OF LOW TO
MODERATE ORGANOPHOSPHATE
EXPOSURE IN LAWN AND LANDSCAPE
APPLICATORS**

Amy E. Brown, Doctor of Philosophy, 1993

**Dissertation directed by: Dr. Robert E. Menzer, Professor Emeritus,
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Exposure to organophosphates (OPs) has been associated with behavioral manifestations including mood changes; reduction of psychomotor speed; and impairment of concentration, memory, language function, and information processing. These effects have been demonstrated in case reports and in tests administered at a single point in time in association with prior high exposure to relatively high doses. However, most applicators are in fact exposed to relatively low levels of insecticide over a prolonged period. The present study was designed to assess the effects of low to moderate OP exposure throughout a spray season. Cholinesterase levels (both plasma and red blood cell) and neurobehavioral effects were investigated. Pre-, mid-, and postseason tests were administered, allowing subjects to serve as their own controls, and results were compared with those of a separate control group of nonexposed workers.

Behavioral impairment was not demonstrated in this study. Thus, at the exposure levels experienced by subjects in this study, OPs do not appear to alter the functions measured (perception, motor reflexes, memory, and affect).

**NEUROBEHAVIORAL EFFECTS OF LOW TO MODERATE
ORGANOPHOSPHATE EXPOSURE IN LAWN AND LANDSCAPE
APPLICATORS**

by

Amy E. Brown

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DEDICATION

To my husband, Dr. Kenneth M. Brown,
whose encouragement and support
made this dissertation a reality

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Introduction

Background

Organophosphate (OP) insecticides are used in the United States and worldwide on many field, fruit, and vegetable crops; in greenhouses, forests and marshes; inside structures; and on home landscapes. Their use has led to cases of poisoning through both occupational exposure and, infrequently, incidental exposure through consumption of treated foods or accidental exposure. Occupational exposures to OP pesticides may occur during manufacture, distribution, application, or reentry. Most OPs are available for use without special training, and it is not uncommon to find them handled by homeowners or by unskilled, temporary laborers. Russell and Overstreet (1987) cite a figure of up to 500,000 persons worldwide requiring clinical attention each year due to OP insecticide exposure. California is the only state that currently requires medical monitoring of workers exposed to OPs (Coye *et al.*, 1986). Eskenazi and Maizlish (1988) argue that agricultural workers may face an elevated risk of mortality from accidents caused, at least in part, by a decrease in neurobehavioral function and from neuropsychiatric diseases.

Metabolism of OPs

OPs are readily absorbed through the skin, lungs, gastrointestinal tract, and conjunctiva. Most are able to pass placental and blood-brain barriers. Oral rat LD₅₀ values of technical ingredients of products on the market in the United States range from 1 mg/kg to over 1400 mg/kg (Fairchild, 1977). Differences in potency are due to different side chains, leading to alternate metabolic pathways, changes in degradation and/or clearance rates, and differences in affinity for toxicity endpoint receptors. OPs can affect both the central and peripheral nervous systems.

Metabolism of OPs can result in intermediate products that are either more or less toxic. These metabolic reactions are classified as activative in the former case and degradative in the latter (O'Brien, 1967). The major activation reactions are oxidative desulfuration and thioether oxidation (Costa, 1988). These reactions are catalyzed by the mixed-function oxidases; the coenzyme nicotinamide adenine dinucleotide phosphate (NADPH) and molecular oxygen are necessary cofactors for the reactions. Many biochemical reactions are involved in a reduction of the toxicity of OPs. The main degradative reactions are: 1.) oxidative O-dealkylation, 2.) oxidative dearylation, 3.) phosphotriester hydrolysis, 4.) hydrolysis of functional groups, and 5.) glutathione transferase reactions (Costa, 1988). Most metabolism of OPs takes place in the liver, although it has been shown to occur in lung and brain tissue (Norman and Neal,

1976). OPs are metabolized in humans relatively rapidly, and excretion of end products is mainly in the urine.

Cholinesterase inhibition

The most extensively studied adverse effect of OPs is cholinesterase (ChE) inhibition, which is the primary mode of toxic action for this class of compounds. OPs form irreversible complexes with carboxylic ester hydrolases, including butyrylcholinesterase (BChE), found in human plasma and liver, and acetylcholinesterase (AChE), found in human erythrocytes, brain, nerves, synapses, and skeletal muscle (Namba, 1971; Silver, 1974). In general, the ChE present in the sera of vertebrates is BChE. Serum ChE is secreted primarily by the liver (Silver, 1974; Schuman, 1986). These esterases are responsible for the hydrolysis of butyrylcholine (BCh) and acetylcholine (ACh). ButyrylChE has also been referred to as plasma ChE (Silver, 1974) and pseudoChE (Mendel and Rudney, 1943), while AChE has been called true cholinesterase (Mendel and Rudney, 1943).

AChE is a glycoprotein which exists in asymmetric (A) and globular (G) forms (Brimijoin, 1983; Kerkut, 1984). The latter may exist as monomers, dimers, or tetramers. The monomer has a molecular weight of approximately 80,000 daltons (Rosenberry and Richardson, 1977; Massoulie *et al.*, 1980). The A forms have a triple-stranded tail of a collagen-like substance attached to one

(A₄), two (A₈), or three (A₁₂) molecules of the G tetramer (Anglister and Silman, 1978; Rosenberry *et al.*, 1980). All forms of AChE appear to be catalytically equivalent (Vigny *et al.*, 1978). In a comparison of human and rat AChE, all molecular forms reacted in the same manner with the respective AChE antibodies, suggesting that all AChE within a particular species may be derived from the same multigene family, and that differences may be due to posttranslational modifications (Marsh *et al.*, 1984).

There is evidence to indicate that the G₁ forms may be the precursors of the other molecular forms of AChE. The dimer is theorized to be formed from the monomer by means of a single disulfide bond (Rosenberry and Richardson, 1977; Anglister and Silman, 1978), and the tetramer from two dimers possibly joined by Van der Waals forces (Brimijoin, 1983). Finally, the asymmetric molecules would be formed by adding tetrameric catalytic units to the collagen tails of the globular forms (Brimijoin, 1983). Following inhibition of AChE with antiChE agents, the G₁ molecule was found to exhibit a rapid recovery, while the G₄ form recovered much more slowly, confirming that synthesis of the latter form may be dependent on reconstruction of the former (Rieger *et al.*, 1976; Gisiger *et al.*, 1977; Koenig and Vigny, 1978; Cisson *et al.*, 1981).

AChE is widely distributed in tissues and organs, including nerve synapses, neuromuscular junctions, brain, liver, blood, plasma, optic tissues, and other organs (Silver, 1974). The molecular form present varies with the tissue

and species (Brimijoin, 1983). Globular forms of the enzyme comprise the majority of AChE in most vertebrate tissues (Brimijoin, 1983; Massoulie and Bon, 1982), with almost all of the human forebrain AChE existing as a tetramer (G_4) (Razon *et al.*, 1984). The asymmetric form has been found in small quantities in motor nerves of both rat and humans (Couraud and Di Giamberardino, 1980; Fernandez *et al.*, 1983). The heterogeneity of the forms of AChE affects the rate of overall recovery of activity. Michalek *et al.* (1982) have shown that medium molecular weight forms play an important role in recovery of enzymatic activity in the brain.

Synaptic acetylcholine (ACh) is released from vesicles in cholinergic nerve endings and binds to nicotinic and muscarinic receptors on the postsynaptic membrane. AChE, in close proximity to ACh receptors on the postsynaptic membrane, then hydrolyzes ACh in a 3-step process involving binding to the enzyme, acetylating the enzyme, and reacting with water to produce acetic acid and the reconstituted enzyme. The reaction is completed rapidly, freeing the esterase for the next molecule of ACh.

Circulating OP compounds bind to BChE and AChE (although with differing affinities), resulting in release of the organic moiety of the OP and phosphorylation of the enzymes at the esteratic site. A hydroxyl group on a serine residue at the active site of ChE reacts with the OP (O'Brien, 1967). The overall reaction is represented in Figure 1. For OPs, k_2 is moderately fast while k_3 is very slow, so that the enzyme-inhibitor complex accumulates. Since

AChE is no longer available to hydrolyze ACh, the neurotransmitter saturates the receptors and accumulates in the synapses, causing overstimulation and, later, blockage of further nerve impulse transmission. The rate of dephosphorylation of the esterase-OP complex is so slow that OPs are referred to as irreversible inhibitors. Carbamates, which similarly inhibit ChE through carbamylation of the enzyme, form less stable complexes and are thus referred to as reversible inhibitors. OPs may exhibit a phenomenon referred to as "aging" wherein the inhibited enzyme becomes refractory to reactivation. The amount of phosphorylated enzyme refractory to reactivation increases with time. The rate of aging has been shown to depend on dealkylation of the phosphoryl group (Karczmar *et al.*, 1970). Increased availability of AChE depends almost entirely on its rate of synthesis. Because there is no other mechanism for removal of ACh from the synapse, the process depends entirely upon AChE, leading to rapid accumulation of ACh at cholinergic synapses when the esterase is inhibited. Repeated exposures to OPs, even at relatively small doses, can continue to deplete the level of available esterases. Most obvious signs and symptoms of OP poisoning are attributable to overstimulation of the cholinergic system through inhibition of AChE. Cholinergic transmission is primarily dependent upon nicotinic receptors in the spinal cord, but both muscarinic and nicotinic receptors are utilized by cortical and subcortical areas of the brain (Taylor and Brown, 1989). The cholinergic synapses in which ACh is the transmitter include those of the central nervous system (CNS), neuromuscular

junctions, sensory nerve endings, ganglionic synapses of both sympathetic and parasympathetic nerves, postganglionic sympathetic nerve endings of sweat glands and blood vessels, and all postganglionic nerve endings of the parasympathetic pathway. The majority of receptors in the brain are muscarinic, while nicotinic receptors mediate spinal, ganglionic and skeletal muscle responses. Cholinergic hyperactivity is manifested in excessive motor activity, skeletal muscle paralysis, and, finally, autonomic dysfunction (Barchi, 1989). Signs and symptoms of OP poisoning include increased bodily secretions, lightheadedness, dizziness, headache, nausea, tremulousness, and loss of muscle coordination in early stages; chest pain, abdominal cramps, drop in blood pressure, increasing shortness of breath in moderate stages; and eventually convulsions, coma, and death due to respiratory failure. Symptoms of acute poisoning occur within twelve hours, and usually within four hours, of overexposure (Morgan, 1989). The nicotinic responses are very rapid, with the muscarinic responses requiring comparatively more time (Table 1). In a retrospective study of patients acutely poisoned by OPs, 92% of the cases manifested peripheral muscarinic symptoms, 44% showed peripheral nicotinic effects, and 40% evidenced CNS effects; only 17% showed involvement of all three (Hirshberg and Lerman, 1984; Lerman *et al.*, 1984).

Tolerance to OPs

In the course of an early experiment involving chronic feeding of the OP insecticide parathion to rats, an interesting phenomenon was noted by Barnes and Denz (1951). After two months on the diet, certain signs of toxicity (excessive lacrimation, salivation, and fasciculation) diminished and the rats remained essentially free of these signs for the duration of the study. Several later studies confirmed the development of tolerance to OPs with regard to both physiological and behavioral symptoms (Clark, 1971; Bignami *et al.*, 1975; Russell *et al.*, 1975; Costa *et al.*, 1982). Tolerance is characterized by an increased sensitivity to cholinergic antagonists and a decreased sensitivity to cholinergic agonists, while ChE activity levels remain depressed (Costa *et al.*, 1982). The development of tolerance is a homeostatic adaptive response which allows the organism to avoid toxic manifestations while under continued toxic insult (Costa, 1992).

Tolerance to OPs is believed to be due in large part to down-regulation, a reduction in the density of ACh receptors, particularly the muscarinic type, in the brain and peripheral organs (Russell *et al.*, 1975; Ehlert, 1980; Costa *et al.*, 1981; Costa and Murphy, 1982; Costa and Murphy, 1983; Lim *et al.*, 1986). Prolonged exposure to acetylcholine and agonists results in a reduction in ACh receptors. Nicotinic receptors, although affected, appear to be less sensitive to OPs (Russell *et al.*, 1975, Costa *et al.*, 1982). The down-regulation of muscarinic

receptors is believed to be due to reduced synthesis of the receptors at the transcription level (Jett and Hill, 1993). Down-regulation of muscarinic receptors takes place fairly slowly and is theorized to be mediated by second messengers created by phosphoinositide hydrolysis (El-Fakahany and Cioffi, 1990). Russell and Overstreet (1987) cite other possible contributing factors to tolerance development, such as the uncoupling of receptors from the generation of their second messengers, potential involvement of other neurotransmitters, and a behaviorally augmented component.

Workers exposed to the high concentrations used in most of the animal studies on tolerance would most likely develop symptoms of toxicity and, therefore, stop, at least temporarily, their exposure to the compounds. However, pressure from employers, themselves (many pesticide applicators are self-employed), masking of signs and symptoms of toxicity (applicators often attribute possible early signs of OP poisoning such as dizziness, nausea, cramps, and excessive sweating to the heat or to other illnesses), or failure to develop symptoms of toxicity could lead applicators to continue their exposure and thus ultimately develop tolerance. Several studies have shown that tolerance can develop in rats in the absence of signs of acute toxicity through repeated low dose exposure (Chippendale *et al.*, 1972; Schwab and Murphy, 1981; Bushnell *et al.*, 1991).

CNS effects

Different areas of the brain display marked differences in sensitivity to OP exposure. Russell *et al.* (1975) found that when the level of whole brain activity of AChE was reduced to about 30% of normal, the ACh level was elevated to about 140%. Yamada *et al.* (1983a,b) have reported that the decrease in muscarinic ACh receptors is region-specific. Decreases were observed in the cerebral cortex, hippocampus, striatum, and ile um, but not in the hypothalamus, thalamus, or midbrain. They suggested that the variations follow the differences in density of cholinergic innervation of the different brain regions. Other researchers have had similar results (Churchill *et al.*, 1984a,b; Shih *et al.*, 1986; Veronesi *et al.*, 1990).

Degeneration of nerve fibers in the brain has been observed following administration of OPs. Degeneration begins at the distal ends of small myelinated long fibers in both ascending and descending tracts of the gray matter (Ecobichon, 1982). Veronesi *et al.* (1990) offered the theory that long-term AChE inhibition may produce a persistent depolarization from the cholinergic septal afferents, resulting in metabolic exhaustion and, eventually, degeneration of target neurons. It has also been suggested that OPs may cause alterations in cerebral blood flow resulting in hypoxia (Aoyagi *et al.*, 1975).

Direct action on ACh receptors

Recently, OPs have been shown to be capable of direct but reversible action on some ACh receptors, particularly of the muscarinic type (Jett *et al.*, 1991; Eldefrawi *et al.*, 1992). The M₂ subtype is the most sensitive to OPs (Bakry *et al.*, 1988) and is present in mammalian brain and cardiac muscle. The M₂ subtype has a high affinity for some OPs (Silveira *et al.*, 1990). OPs exhibit poor correlation between the degree of AChE inhibition and their affinities for the M₂ receptor (Eldefrawi *et al.*, 1992), which may help to account for the poor correlation between rat oral toxicity and degree of AChE inhibition noted by Chambers *et al.* (1990).

Delayed neuropathy

A rare type of polyneuropathy characterized by delayed onset has been caused by several non-insecticidal OPs such as tri-*o*-cresol phosphate (TOCP) and by some insecticides, of which chlorpyrifos is the only such insecticide marketed in the U.S. (Lotti *et al.*, 1986; Namba *et al.*, 1971). Symptoms typically begin two to four weeks after a single sufficiently large exposure to the agent (Karalliedde and Senanayake, 1989) and are characterized by weakness of variable severity of distal limb muscles, especially in the lower limbs, accompanied by cramping, tingling, and ataxia. This type of neuropathy is

classified as central-peripheral axonopathy, involving degeneration of long and large diameter fibers (Richardson, 1985; Lotti *et al.*, 1986). The syndrome is not correlated with ChE depression, but is associated with a high level of phosphorylation of the nervous system enzyme, neuropathy target esterase (NTE), sometimes also referred to as neurotoxic esterase (Lotti and Johnson, 1978; Lotti *et al.*, 1981; Lotti *et al.*, 1986). Although inhibition of the enzyme takes place within hours after exposure, the events leading to the onset of polyneuropathy have not yet been identified. It has been shown, however, that NTE activity in the nervous system is usually fully restored by the time the syndrome is manifest (Lotti *et al.*, 1986).

Behavioral effects

The down-regulation of muscarinic ACh receptors observed in individuals exposed to anticholinesterase agents over long periods, while providing some protection against acute toxicity, may result in undesirable effects on the brain. Cholinergic mechanisms have been shown to be important in memory (Deutsch, 1973; Moss and Deutsch, 1975; Davis and Yamamura, 1978), in mental disorders and various neuropsychiatric syndromes (Weiss *et al.*, 1976; Davis *et al.*, 1978), and sleep disorder (Jouvet, 1975). Several higher brain functions, including memory, learning, sensorimotor functioning, and attention have shown impairment when cholinergic transmission is blocked with muscarinic antago-

nists (Douglas and Isaacson, 1966; Costa and Murphy, 1982; McDonald *et al.*, 1988; Eckerman and Bushnell, 1992). Recently, Riekkinen *et al.* (1990) have shown an interaction of nicotinic and muscarinic receptors in regulation of performance of certain spatial and learning tasks.

Exposure to OPs has been associated with behavioral changes in humans by several investigators. OP-exposed individuals have exhibited increased anxiety, confusion, depression, impaired concentration and memory, language function effects, and changes in electroencephalogram (EEG) patterns (Grob *et al.*, 1947; Rowntree *et al.*, 1950; Gershon and Shaw, 1961; Dille and Smith, 1964; Bowers *et al.*, 1964; Holmes, 1964; Durham *et al.*, 1965; Redhead, 1968; Smith *et al.*, 1968; Metcalf and Holmes, 1969; Simson, 1971; Namba, 1971; Drachman and Leavitt, 1974; Janowsky *et al.*, 1974; Levin *et al.*, 1976; Ecobichon *et al.*, 1977; Korsak and Sato, 1977; Duffy *et al.*, 1979; Hirshberg and Lerman, 1984; Savage *et al.*, 1988; Rosenstock *et al.*, 1991; Table 2.). Behavioral changes and EEG abnormalities have been shown to persist long after recovery of baseline ChE levels (Duffy *et al.*, 1979; Whorton and Obrinsky, 1983; Savage *et al.*, 1988; Gralewicz *et al.*, 1990) and have also been noted in the absence of any significant depression of ChE. These effects may be more sensitive to nervous tissue dysfunction resulting from long term exposure to low doses of OPs, such as is typical of many of the workers who apply these compounds in pest control businesses. Tabershaw and Cooper (1966) found approximately 15% of cases with persistent neurobehavioral effects had

repeated or multiple exposures. Silbergeld (1983) suggested that neurobehavioral evaluations should be used for monitoring exposure to all toxic chemicals, as they often present earlier signs of poisoning than clinical symptoms.

In studies with laboratory animals, impaired behaviors followed continuously throughout exposure to OPs generally returned to normal as exposure continued, indicating the development of tolerance to the antiChE agent. The time course of recovery, however, was dependent on the particular behavior under investigation (Overstreet, 1984). Raffaele *et al.* (1990) reported that even when OP exposure by itself produced no detectable difference in behavior, changes in the brain made behavior more susceptible to additional stress on the system. In a study of rats injected with the OP soman, it was found that ChE activity in the brain correlated poorly with both plasma and RBC ChE activity, moderately with behavioral scores, and well with spinal cord activity (Jimmerson *et al.*, 1989). In general, significant recovery of AChE activity levels preceded the return to normal behavioral performances. Early studies had suggested that a threshold decrease of 40% of brain AChE activity was necessary before cholinergically mediated behavior changes would be manifested (Russell *et al.*, 1961, Glow and Rose, 1965). However, Kozar *et al.* (1976) demonstrated that some functions may recover before AChE activity returned to 40% of normal, suggesting that other factors, such as adaptation of postsynaptic cells to AChE, resynthesis of various AChE isozymes or increased activity of alternative

noncholinergic systems, may play important roles in recovery of behavioral functions.

Case reports of persons poisoned by various OP insecticides gave the first indications of possible behavioral effects in humans (Gershon and Shaw, 1961; Holmes, 1964; Dille and Smith, 1964; Smith *et al.*, 1968; Redhead, 1968; Simson *et al.*, 1969; Namba, 1971; and Ecobichon *et al.*, 1977). Several early reports came from accidents involving agricultural pilots, or "crop dusters." It was common practice at the time for the same individual to mix the concentrate pesticide into the desired dilution, load it into the spray tanks, and then pilot the plane. Even mild symptoms such as confusion contributed to aircraft accidents, often exposing the pilot further to the compound and resulting in more lasting effects (Dille and Smith, 1964; Smith *et al.*, 1968; Clark, 1971; Namba, 1971). Psychiatrists investigating effects associated with the clinical administration of OP drugs to patients under treatment for mental illness also provided case studies (Rowntree *et al.*, 1950; Drachman and Leavitt, 1974). After the initial case reports raised questions, a number of studies were designed to investigate the association between OPs and behavioral changes under more controlled circumstances. Results varied and were sometimes contradictory. Interpretation of results and, especially, comparison of studies, is complicated by the lack of standardization of tests to assess neurobehavioral function, lack of exposure estimates, occasional inappropriate (or lack of) controls, and incomplete reporting of methodologies.

In the course of investigating the causes of depression, valuable information was gained from studies involving the OPs scopolamine, its congener methscopolamine, physostigmine, and neostigmine. Scopolamine and physostigmine act both centrally and peripherally, while methscopolamine and neostigmine act only peripherally as they do not cross the blood-brain barrier easily. Thus, methscopolamine acts as a matched peripheral control for the behavioral effects of scopolamine, and neostigmine for physostigmine (Drachman and Leavitt, 1974), allowing the contributions of peripheral actions on centrally-mediated behavior to be factored out. In a study of mental patients, Janowsky *et al.* (1974) found that virtually all patients receiving physostigmine exhibited symptoms of psychomotor retardation, which was attributable solely to the CNS effects of the drug, as the symptoms did not appear with the administration of neostigmine. Drachman and Leavitt (1974) determined that scopolamine significantly affected aspects of memory and cognitive functioning, but methscopolamine did not.

Rodnitzky *et al.* (1975) administered neurobehavioral tests to a group of farmers and commercial pest control applicators who had applied OPs within the prior two weeks and compared the results to those of farmers either tested prior to the beginning of the spray season or who had not been involved in spray operations. The mean plasma ChE level of the exposed group was significantly lower than that of the control group; RBC ChE levels were not significantly different. Occupationally exposed subjects performed equally as

well as controls on each of the tests, designed to assess memory, signal processing, vigilance, language, and proprioceptive feedback performance. Levin *et al.* (1976) reported further results from this study, indicating highly significant differences in anxiety (pest control operators exhibited more anxiety than their controls, but farmers did not), but no differences in mental depression. Factors that might well have affected this study include the early testing of some control subjects compared to exposed, and the use of pest control operators who spray different pesticides at different concentrations indoors and thus have a very different exposure profile than farmers.

Thirty-two male subjects occupationally exposed to OPs were tested over a four-hour period by Korsak and Sato (1977). An index was devised to determine the level of chronic exposure to OPs, and the subject pool was subsequently divided in half, comparing values for those who rated in the upper half of the exposure index against those who rated in the lower half. Plasma ChE did not differ between the two groups; RBC ChE was not measured. Significant differences were found on the Bender Visual Motor Gestalt Test and on Part B of the Trail Making Test, with subjects scoring in the upper half of the index exhibiting poorer performance, and on EEGs. None of the other behavioral effects tested (tactual performance, memory, localization, finger tapping, and Zung Self-rating Depression Scale) showed significant differences.

Maizlish *et al.* (1987) investigated the effects of short-term low-level occupational exposure to the OP diazinon. Forty-six applicators were tested

against fifty-three supervisors and inspectors not involved in spraying. Before and after the work shift, urine was analyzed for OP metabolites and tests from the NES battery (see section on measurement of behavioral effects) were administered. The results showed minor performance impairment in the exposed group; however, a dose-related effect was not demonstrated. The granular formulation used contains only 14% diazinon, and is considered to have a very low probability of absorption by the applicator unless the granules become wet, releasing the active ingredient. The normal amount of protective gear required by the product labeling includes long-sleeved pants, a long-sleeved shirt, shoes and socks, and chemical-resistant gloves. However, subjects exposed to diazinon in this experiment were required to wear disposable protective coveralls, rubber boots, and rubber gloves while in the treatment area, as well as a full face shield and an air-purifying (cartridge type) respirator while loading the insecticide into the spreader. Therefore, the applicators in this study would be expected to have absorbed extremely low, if any, amounts of diazinon during the test period. The authors cited the low dose level, short duration of exposure, beginning of testing after applicators had already been exposed for one month, differences in the frequency of urine sample collection, and use of a time frame for urine collection that was "probably shorter than that for urinary clearance" as limitations of the study. NES performance of both exposed and non-exposed subjects improved on the post-shift tests compared to pre-shift, indicating a learning effect and complicating analysis of results. In

addition, non-exposed subjects reported more frequent non-occupational use of pesticides than the exposed group.

Savage *et al.* (1988) administered a battery of tests to subjects with at least one documented OP poisoning incident occurring 35 to 10 years prior to testing, and demonstrated chronic neurological sequelae to acute poisoning. These effects, detected through the use of subtests from the Halstead-Reitan Battery, the Peabody Individual Achievement Test, the Wechsler Adult Intelligence Scale, and the Minnesota Multiphasic Personality Inventory, were sufficiently subtle that the clinical neurological examination could not detect differences between the exposed and control groups.

Rosenstock *et al.* (1991) compared neuropsychological function in a group of agricultural workers approximately two years after OP poisoning to that of a matched control group and found differences between the groups with respect to auditory attention, visual memory, visuomotor speed, sequencing and problem solving, and motor steadiness, reaction, and dexterity. The results of this study are consistent with those of Savage *et al.* (1988).

Two of the previous studies also investigated the effect of organochlorine insecticide exposure on neurobehavioral performance. Although there is no direct evidence for such effects, the relatively long persistence of organochlorines in body tissues makes it prudent to investigate this possibility. When Korsak and Sato (1977) classified their subjects on the basis of high versus low serum residues of dieldrin, they found no significant difference in neurobe-

havioral functions tested between the two groups. The OP-exposed subjects in the study conducted by Savage *et al.* (1988) had a significantly higher mean serum level of organochlorine residues than controls. However, an analysis of covariance failed to show any significant association between neurobehavioral effects and organochlorine residues. Thus exposure to organochlorines, and the continued presence of residues in body tissues, appear to be unimportant with regard to behavior.

Measurement of exposure: ChE levels

Chronic exposure to OPs can produce functional changes in the nervous system that may not be detectable without an additional challenge, but may have serious consequences in response to such a challenge (Raffaele *et al.*, 1990). Therefore, ChE activity is often monitored to determine the exposure level of a worker to OPs. Blood ChE activity is commonly used as an indirect estimator of the degree of inhibition of synaptic ChE. Both erythrocyte (RBC) ChE and plasma ChE can be measured. AChE is present in human erythrocytes and is used to provide an estimate of the effects occurring in the target organs, assuming the antiChE agent has equal access to blood and synapses (Karalliedde and Senanayake, 1989). Although plasma ChE does not accurately reflect AChE activity of tissues, it is considered a very sensitive indicator of exposure to OPs (Peedicayil *et al.*, 1991; Namba *et al.*, 1971). RBC ChE is less

likely than plasma ChE to be affected by factors other than OPs, including pathologic conditions (advanced liver disease, malnutrition, and dermatomyositis), other toxicants (carbon disulfide, benzalkonium salts, organic mercury compounds, ciguatoxins, and solanines), and hormonal conditions (such as occur in early pregnancy or with birth control pills) (Morgan, 1989). In a recent review, the U. S. Environmental Protection Agency (EPA) Scientific Advisory Board (SAB)/Scientific Advisory Panel (SAP) Joint Study Group expressed doubt whether plasma and RBC ChE inhibition are valid indicators of toxicity, but unanimously recommended that ChE inhibition continue to be treated as a useful measure of exposure to OPs (U. S. EPA, 1990).

Manifestations of acute OP poisoning occur when more than 50% of plasma or RBC ChE is inhibited, with severity of symptoms in the initial stage of acute poisoning well correlated with the degree of inhibition. Levin and Rodnitzky (1976) found EEG abnormalities to be positively correlated with the level of AChE inhibition during initial stages of intoxication. Standard degrees of inhibition of plasma ChE associated with different levels of poisoning have been reported as 20 to 50% inhibition in mild poisoning, 50 to 90% in moderate poisoning, and over 90% inhibition in severe poisoning (Grob and Harvey, 1953; Namba *et al.*, 1971; Karalliedde and Senanayake, 1989). Clinical recovery occurs by the time RBC ChE levels have returned to 30% of normal (Kipling and Cruickshank, 1985). ChE activity in the blood has been found to normalize faster than in some parts of the brain (Gralewicz *et al.*, 1990). The

rate of recovery of ChE activity varies with the insecticide (Yeary, 1984). Plasma ChE exhibits depression prior to RBC ChE and is thus a more sensitive indicator of exposure. However, plasma ChE levels also return to normal more quickly, requiring about 30 to 40 days (Levin *et al.*, 1976), with about 13% restoration achieved within 24 hours (Durham and Hayes, 1962). RBC ChE inhibition may not reach its maximum for several days and recovery is dependent upon replacement of erythrocytes in the blood, i. e., 90 to 100 days (Levin *et al.*, 1976). Daily exposure to amounts of OPs insufficient to cause signs and symptoms of acute poisoning may result in a chronic illness characterized by weakness, anorexia, and malaise (Morgan, 1982). Depression of ChE activity associated with this condition is cumulative and shows no predictable correlation between the degree or severity of symptoms and the degree of either plasma or RBC ChE inhibition. Subsequent exposure to low levels of OPs, however, may result in a recurrence of acute symptoms (Dille and Smith, 1964).

Several studies have analyzed the interindividual differences in ChE activity, attributing variations to physiological factors such as sex, body weight, height, and ChE-1 phenotype (Sidell and Kaminskis, 1975; Lepage *et al.*, 1985; Moses *et al.*, 1986; Brock and Brock, 1990). These studies also describe significant intraindividual variation independent of age, weight, height, sex, and ChE-1 phenotype. Seasonal variation in ChE activity also exists, with lower values detected during the summer months (Yeary, 1984). Brock and Brock (1990) found that the potential analytical variability of the ChE assay is

insignificant compared with the physiological variability. Therefore, establishment of a worker's own baseline ChE level is very important when monitoring, as the normal range of values is wide. Currently, 25% to 30% reduction of ChE is widely accepted as the threshold at which a person should be prevented from further exposure to OPs until the level returns to baseline (Yeary, 1984; Cassidy, 1989; Brock, 1991).

Dow Chemical USA reported the results of several studies on ChE depression over time as a result of exposure to the insecticide, chlorpyrifos. In a small study of female volunteers exposed to dermal concentrations for twelve hours at a time with twelve hours between exposures, two of the volunteers showed no depression of either plasma or RBC ChE relative to their pretreatment levels (Kilian *et al.*, 1970). One subject was exposed to 25 mg chlorpyrifos/kg body weight for two test periods, the other to 10 mg/kg for four test periods. A third subject displayed plasma ChE values of 67% and 47.5% of her pretreatment level at 12 and 60 hours respectively following exposure to 25 mg/kg for three test periods; her RBC ChE was unaffected. A fourth subject exposed to 5 mg/kg for twenty test periods showed no significant plasma ChE depression after the fifth, tenth, or fifteenth period, but exhibited a depression to 64.3% of her pretreatment level after the twentieth exposure. RBC ChE values were unaffected. In a field study conducted by the U. S. Public Health Service (USPHS), ChE levels of men who applied Dursban M as a 0.5% chlorpyrifos spray emulsion using high pressure spray systems were

monitored. One of seven men exhibited a 90% depression of his pretreatment level of plasma ChE; the others showed a 50% decrease. RBC ChE was unaffected in all seven (Kenaga, 1967; Eliason *et al.*, 1969). No information was given in these reports concerning how long after spraying the ChE sampling was conducted. In another USPHS study, ChE levels of seven men applying a concentration of chlorpyrifos approximately five times greater than the recommended rate were monitored (Eliason *et al.*, 1969; Schoof, 1968). Four of the subjects showed a depression of plasma ChE ranging from 41% to 91%. Four days after the application, a fifth subject showed 53% depression in plasma ChE. The remaining two exposed subjects showed lesser depreciation (7% and 15%) but had applied a significantly smaller volume of spray and thus had less exposure. ChE levels determined 17 and 37 days posttreatment showed a gradual return to pretreatment values except for one subject whose ChE level returned only to 75% of his pretreatment level. Again, RBC ChE levels were unaffected.

The standard test for ChE level is based on the colorimetric method devised by Ellman *et al.* (1961; Figure 2). The assay is based on the reaction of the color reagent 5,5-dithiobis-2-nitrobenzoic acid (DTNB) with the thiocholine which is released upon hydrolysis of the substrate, such as acetylthiocholine iodide, in the presence of a ChE. The reaction of DTNB with the thiocholine is instantaneous, thus the rate limiting factor when sufficient substrate is available is the enzyme concentration. The yellow product, with an

absorbance maximum of 412 nm, is analyzed spectrophotometrically to determine the number of moles of thiocholine ester substrate hydrolyzed. One unit of activity is defined as the amount of ChE necessary to hydrolyze 1 μ M of substrate per minute at 37⁰ C. Intraindividual variability of RBC ChE as measured by the coefficient of variation has been reported as 10.3% and 10.0%, while the corresponding value for plasma ChE has been reported as 10.3% and 14.4% (Gage, 1967; Yager *et al.*, 1976). Variability due to interindividual differences was determined to be 17.0% and 15.1% for plasma ChE, and 7.0% and 8.5% for RBC ChE by these authors. Most assays currently run, both in experimental and clinical cases, use a modified Ellman methodology. Each method has its own "normal" range (Karalliedde and Senanayake, 1989), thus care must be taken to use the same methodology for monitoring any individual.

Other assays for detection of ChE activity include electrometric, titrimetric, radiometric, fluorometric, gasometric, and immunochemical methods. However, the Ellman assay remains the method of choice because it is easily adaptable to automation and is thus more reproducible, more precise, and more easily performed (Chin, 1992).

Measurement of exposure: Behavioral effects

Neurobehavioral tests have been widely used for many years to evaluate health effects following exposure to neurotoxic agents (Johnson and Anger,

1983). During a 1983 meeting convened by the World Health Organization (WHO) and the U. S. National Institute for Occupational Safety and Health (NIOSH), a call was issued for standardized tests (Anger *et al.*, 1991). The group proposed a battery of tests, later termed the National Core Tests Battery (NCTB), which could identify or screen for neurotoxic effects of chemicals worldwide. Recently, the Neurobehavioral Evaluation System (NES), a computerized adaptation of nineteen commonly used neurobehavioral tests was developed (Baker *et al.*, 1985a,b; Letz, 1988; Eisen, 1988) and has been applied in epidemiological studies of exposure to various toxic agents (Greenberg *et al.*, 1985; Fidler *et al.*, 1987; Baker *et al.*, 1988; Schmedtje, Jr. *et al.*, 1988; Hooisma *et al.*, 1988; and Mahoney *et al.*, 1988). Computerization has allowed testing where neurobehavioral data collection would not have occurred otherwise, as it increases reliability and efficiency of test administration (Anger, 1992). The NES tests include variants of five of the tests recommended by the WHO/-NIOSH group, and has become the most widely used behavioral test battery for assessment of exposure to neurotoxic chemicals (Anger, 1990; Anger, 1991). In development of the NES, particular emphasis was placed on adaptation of tests that had proved useful in previous occupational studies (Letz, 1988). Test-retest correlations from various laboratory studies utilizing NES ranged from 0.51 to 0.92, with an average of 0.76. Field studies provided values ranging from 0.61 to 0.87, with an average test-retest correlation of 0.70. Stability of the tests, important if they are to be administered repeatedly, was generally good; when

improvement was noted, it tended to occur between the first and second presentations of the tests, suggesting a one-time learning effect (Baker *et al.*, 1985). Sensitivity is another critical consideration of the test method. Greenberg *et al.* (1985) and Mahoney *et al.* (1988) were able to detect effects using a subclinical dose of nitrous oxide, a known neuroactive agent, even though the number of subjects was very small (twelve and fifteen male volunteers, respectively). Maizlish *et al.* (1987) estimated that the NES battery used in their study was sensitive to differences of 5 to 10% between exposure groups. Anger *et al.* (1990) reported that control versus exposed group differences on NCTB tests in populations exposed to lead, mercury, and carbon disulfide have been replicated extensively in independent studies. Anger (1992) lists the simple reaction time, digit span, and finger tapping tests among those tests found to be most sensitive to effects of neurotoxicants reported from laboratory experiments. Under field worksite conditions, simple reaction and finger tapping were again among the most sensitive, as were the digit-span and symbol-digit tests. Chrislip (1991) found that males and females performed similarly on most tests in both the NCTB and NES batteries. When differences did occur they were relatively small and not statistically different.

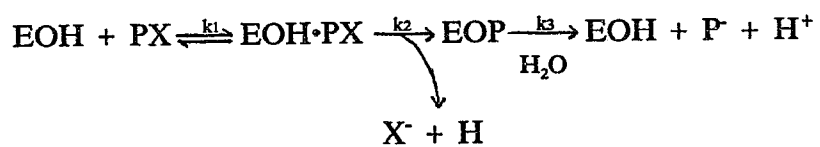
Rationale for the study

The weight of the evidence supports the conclusion that OPs are capable of causing behavioral changes in humans, particularly in those exposed to doses large enough to cause clinical poisoning. While Maizlish *et al.* (1987) attempted to investigate the neurobehavioral effects of low-dose exposure, their study had many confounding factors and, in any case, only examined the applicators at one point in time. No study to date has examined applicators repeatedly throughout the entire period of application. Most applicators are not poisoned but are exposed to relatively low levels of OPs throughout a prolonged period of time. Consumers are exposed to even lower doses of OPs through their food and through intermittent contact with treated surfaces, but concerns have been raised as to the safety of OPs with regard to even these small exposures. Therefore, this study was designed to investigate whether changes in neuropsychological function might occur from such exposures.

The objective of this study was to determine whether applicators exposed to low to moderate levels of OPs throughout a season exhibited differences in neuropsychological function and, if so, whether these differences disappeared after the season ended and exposure stopped. This study differs from those conducted previously by other researchers in the following ways: 1.) exposure histories were developed; 2.) ChE tests were performed to document physiological exposure; 3.) pre-exposure parameters (both ChE levels and performance

on neurobehavioral tests) were collected; 4.) neurobehavioral tests chosen were drawn from the NES, which is concurrently being used in several other studies of occupational exposures, making comparison of results between studies easier; and 5.) tests were repeated throughout the season and once after the season had ended. The inclusion of pre-exposure and post-exposure tests allowed exposed subjects to act as their own controls in addition to having a separate control group of nonexposed subjects.

Figure 1.--OP interaction with AChE¹



EOH = ChE

PX = OP

EOH·PX = reversible complex

¹Adapted from Murphy, 1986.

Table 1.--Peripheral and CNS effects of cholinesterase inhibitors¹

1. Peripheral actions -- muscarinic receptors

Exocrine glands:	Increased salivation, lacrimation, perspiration
Eyes:	Miosis (pinpoint and nonreactive), ptosis, blurring of vision, conjunctival injection, "bloody tears"
Gastrointestinal tract:	Nausea, vomiting, abdominal tightness, swelling and cramps, diarrhea, tenesmus, fecal incontinence
Respiratory tract:	Excessive bronchial secretions, rhinorrhea, wheezing, edema, tightness in chest, bronchospasms, bronchoconstriction, cough, bradypnea, dyspnea
Cardiovascular system:	Bradycardia, decrease in blood pressure
Bladder:	Urinary frequency and incontinence

2. Peripheral actions -- nicotinic receptors

Skeletal muscles:	Muscle fasciculations (eyelids, fine facial muscles), cramps, diminished tendon reflexes, generalized weakness in peripheral and respiratory muscles, paralysis, rigid or flaccid tone
Cardiovascular system:	Tachycardia, pallor, increase in blood pressure

3. CNS actions

Brain:	Drowsiness, lethargy, fatigue, mental confusion, impaired concentration, headache, pressure in head, generalized weakness, tremors, Cheyne-Stokes respiration, dyspnea, convulsions, depression of respiratory centers, cyanosis, coma with absence of reflexes
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¹Adapted from Namba *et al.* (1971), Ecobichon (1982), and Costa (1988)

Table 2.--Behavioral effects associated with exposure to organophosphates¹.

<u>Effect</u>	<u>Author, Year</u>	<u>Result</u>
Confusion/ impaired concentration	Gershon and Shaw, 1961	+
	Dille and Smith, 1964	+
	Bowers <i>et al.</i> , 1964	+
	Durham <i>et al.</i> , 1965	+
	Redhead, 1968	+
	Smith <i>et al.</i> , 1968	+
	Metcalf and Holmes, 1969	+
	Drachman and Leavitt, 1974	+
	Rodnitzky <i>et al.</i> , 1975	--
	Savage <i>et al.</i> , 1988	+
	Hirshberg and Lerman, 1984	+
Information processing and psychomotor speed	Bowers <i>et al.</i> , 1964	+
	Durham <i>et al.</i> , 1965	--
	Metcalf and Holmes, 1969	+
	Namba <i>et al.</i> , 1971	+
	Janowsky <i>et al.</i> , 1974	+
	Rodnitzky <i>et al.</i> , 1975	--
	Korsak and Sato, 1977	+
	Maizlish <i>et al.</i> , 1987	--
	Savage <i>et al.</i> , 1988	+
	Rosenstock <i>et al.</i> , 1991	+
Memory impairment	Gershon and Shaw, 1961	+
	Bowers <i>et al.</i> , 1964	+
	Metcalf and Holmes, 1969	+
	Namba <i>et al.</i> , 1971	+
	Drachman and Leavitt, 1974	--
	Rodnitzky <i>et al.</i> , 1975	--
	Korsak and Sato, 1977	--
	Maizlish <i>et al.</i> , 1987	--
	Savage <i>et al.</i> , 1988	--
	Joubert and Joubert, 1988	+
	Rosenstock <i>et al.</i> , 1991	+

Table 2. (cont'd.)

<u>Effect</u>	<u>Author, Year</u>	<u>Result</u>
Expressive language deficits	Bowers <i>et al.</i> , 1964	+
	Simson <i>et al.</i> , 1969	+
	Namba <i>et al.</i> , 1971	+
	Rodnitzky <i>et al.</i> , 1975	--
	Maizlish <i>et al.</i> , 1987	--
	Savage <i>et al.</i> , 1988	+
Depression	Rowntree <i>et al.</i> , 1950	+
	Gershon and Shaw, 1961	+
	Bowers <i>et al.</i> , 1964	+
	Namba <i>et al.</i> , 1971	+
	Janowsky <i>et al.</i> , 1974	+
	Levin <i>et al.</i> , 1976	--
	Hirshberg and Lerman, 1984	+
	Savage <i>et al.</i> , 1988	+
Anxiety and irritability	Joubert and Joubert, 1988	+
	Gershon and Shaw, 1961	+
	Bowers <i>et al.</i> , 1964	+
	Dille and Smith, 1964	+
	Metcalf and Holmes, 1969	--
	Namba <i>et al.</i> , 1971	+
	Levin <i>et al.</i> , 1976	+
	Ecobichon <i>et al.</i> , 1977	+
	Savage <i>et al.</i> , 1988	+
EEG changes	Joubert and Joubert, 1988	+
	Grob <i>et al.</i> , 1947	+
	Holmes, 1964	+
	Metcalf and Holmes, 1969	+
	Korsak and Sato, 1977	+
	Duffy <i>et al.</i> , 1979	+
	Hirshberg and Lerman, 1984	+
	Savage <i>et al.</i> , 1988	--

¹Note: + indicates effect was looked for and detected; -- indicates effect was looked for and not detected.

Materials and Methods

Outline of the Study

Lawn and landscape OP applicators and nonexposed subjects who perform similar work were solicited to cooperate in the study, which was run from February through November of 1991. Exposure level was objectively measured by ChE tests administered five times throughout the year: once before the spray season began, three times during the spray season, and once after spraying had stopped. A battery of neurobehavioral tests was administered on the same dates. Test sessions were scheduled over a period of two weeks, Monday through Thursday, at one of seven locations chosen for convenience to the particular group of subjects scheduled for that date. Subjects reported to their test facility before beginning work (6:45 - 7:00 a.m.) to have their blood drawn for the ChE analysis. After finishing work, the subjects returned to the same facility to take a battery of neurobehavioral tests lasting approximately 35 minutes.

Review of the methodology

The study protocol was reviewed by the University of Maryland human subjects Institutional Review Board (IRB) in January 1990 and by an expert panel of scientists from the National Institute for Occupational Safety and Health (NIOSH) and the Centers for Disease Control (CDC) at the NIOSH facility in Cincinnati, Ohio in September 1990. The panel was convened "to solicit individual recommendations regarding behavioral and neurochemical testing methods applicable to the study of the neurobehavioral function of organophosphorus pesticides" (Hilyer, 1990).

Selection of subjects

Ninety-seven lawn and/or landscaping firms listed in the 1989-1990 Maryland Directory of Pesticide Licensees and Permittees (Setting, 1990) as registered in Montgomery, Prince George's, and Howard counties, Maryland, were contacted by telephone to determine interest in cooperating in the project. The firms were informed that the project did not involve additional exposure to any chemicals other than those applied in the course of the firm's normal business operations, and that each subject who completed all five sessions of the project would be paid \$100. Employees who applied OPs would be eligible to serve as exposed subjects; employees who performed similar outdoor work but

did not apply OPs could act as controls. In addition, the investigator contacted the Patuxent Wildlife Research Center Facility Management Branch, the Maryland-National Capital Parks and Planning Commission, and a grass seeding company to solicit additional control subjects. All of the firms and agencies were asked questions about their 1990 level of OP use, number of employees spraying OPs throughout a typical season, number of employees performing similar work outdoors but not using OPs, earliest probable date of OPs, and the date by which any seasonal applicators would be hired (Appendix A). A total of 18 firms expressed interest in participating, with a pool of 77 exposed subjects (OP applicators) and sixty-four control subjects (persons whose jobs a.) required them to perform similar work but b.) did not include OP application). The number of potential participants per firm ranged from 1 to 22.

Cholinesterase testing

The investigator contracted with National Health Laboratories (NHL) in Herndon, Virginia to perform ChE analyses. On each test date, a registered phlebotomist employed by NHL traveled to each facility to draw two 7ml tubes of blood from each subject. The samples were kept on ice and were transported back to NHL by the phlebotomist for analysis of both plasma and RBC ChE. At NHL, approximately 1 ml of blood was centrifuged for 3 - 8 minutes at 3000 rpm. Plasma was separated from the RBC portion; 0.1 ml

RBC was resuspended in 3.0 ml distilled water and hemolyzed by mixing. The plasma sample was diluted 1:9 with distilled water. In a cuvette, 0.01 ml sample was mixed with 1.0 ml Gilford reagent #721416 (4.0 mM propionyl-thiocholine iodide, 0.25 mM 5,5-dithio-bis-[2-nitrobenzoic acid], potassium phosphate buffer, pH 6.8). Absorbance at 405 nm was recorded at 15 seconds and at 45 seconds. The change in absorbance in 30 seconds for water was subtracted from that of the sample and the result was multiplied by the appropriate dilution factor. ChE activity in international units per liter (U/l) was calculated according to the following equation:

$$\text{ChE activity (U/l)} = \frac{\Delta A / 30s \times 10^6 \times 1.01 \times 2}{(14.64 \times 10^3) \times 1 \times 0.01} = \Delta A / 30s \times 13,798$$

Where U = the amount of enzyme that catalyzes the hydrolysis of 1 μ mole of substrate per minute under defined conditions

ΔA = change in absorbance

10^6 = conversion of mole to μ mole

1.01 = total reaction volume in ml

2 = conversion of $\Delta A / 30s$ to $\Delta A / \text{min}$

14.64×10^3 = molar absorptivity of 5-thio-2-nitrobenzoic acid at 405 nm

1 = light path in cm

0.01 = sample volume in ml

Results were returned from the lab by courier, arriving within two to six days after the samples had been drawn. Plasma and RBC ChE levels were recorded and the percent change from baseline calculated. Subjects exhibiting a decrease of 25% or more in either plasma or RBC ChE were notified by telephone immediately and advised to discontinue exposure to OPs and carbamates until further notice. The safety officer of the subject's firm was then notified.

Neurobehavioral tests

Tests were chosen from the revised version of the computerized Neurobehavioral Evaluation System (NES2, version 4.2). These computerized versions are adaptations of well-known tests commonly used in investigating effects of occupational exposure to neurotoxic agents (Hartman, 1988). The NES2 presents the tests, records the subjects' responses, and stores the responses in a data file. Individual tests from the entire battery of NES tests available were chosen based upon 1.) sensitivity to the type of effect noted in previous studies; 2.) suitability for repeated measures design; 3.) recommendations of the CDC/NIOSH panel; and 4.) time necessary to complete the test. To encourage participation, approximate total test administration time was kept below 40 minutes. The tests chosen and the functions they measure are described below according to Baker *et al.* (1985b).

- a. The finger tapping test measures motor quickness and accuracy. The subject must press a button as many times in succession as possible over a specified interval. Administration requires 3 minutes.
- b. The continuous performance test is a WHO supplemental test measuring sustained visual attention, and allows indices of speed, task learning, and attention to be devised. The subject is asked to press a button in response to the appearance on the screen of a particular symbol. Various letters are flashed onto the screen at a rate of 1 per second for 5 minutes. Individual response latencies are recorded, as are errors of omission and commission. Time required to administer the test is 6 minutes.
- c. The hand-eye coordination test evaluates dexterity. This test requires the subject to trace a sine wave pattern on the display terminal by using a joystick. The cursor is moved horizontally at a constant rate by the computer, while the subject controls the vertical movement. Deviation from the line is measured by the computer and recorded. Time required to administer the test is 5 minutes.
- d. The symbol-digit substitution test is one of the WHO core tests; it evaluates speed and coding ability. In the NES adaptation, nine symbols and digits are paired on the screen; the subject is then asked to press the digit keys corresponding to the

appropriate symbol. The time required to complete each symbol-digit pairing and the number of incorrect matches are recorded. The time required to administer this test is 5 minutes.

e. The simple reaction time test is a WHO core test; it measures speed of reaction. The subject is asked to press a button in response to the appearance on the screen of a particular symbol. The time between presentations of the stimulus is randomly varied to reduce anticipation effects. Data are recorded as individual reaction times over 60 presentations of the stimulus, and response latencies are averaged over blocks of 12 trials. Time required for administration is 5 minutes.

f. The digit-span test is a WHO core test which evaluates short-term memory. The subject is asked to enter into the computer progressively longer sets of digits which have been presented previously by the computer at a rate of 1 per second. New digit sequences are presented at each span length. After 2 incorrect responses at a span length, the subject is asked to enter the digits in reverse order from that presented by the computer. The test requires 7 minutes to administer.

g. Mood scales are part of the WHO core battery. Subjects are asked to rate themselves with respect to a series of adjectives or phrases on the basis of their feelings during the past week. The NES test presents 25

such choices and yields a 5-dimensional mood profile by combining ratings on individual items. The test requires 5 minutes to administer.

The investigator received training on administering the NES2 tests from two NIOSH neuropsychologists who had extensive experience in use of the tests. Topics covered included: facility set-up, conductance of pretest sessions, encouragement of subjects' effort, monitoring subjects during testing, and data storage.

As each subject entered the test room, he/she was logged in on the computer by the investigator or her assistant. Subjects were reminded each time they began the test battery that they should try as hard as possible on all tests and to proceed as quickly as possible through the test battery. Subjects were observed throughout the test period and any problems were recorded in a log. The tests were run on 6 JVC desktop computers transported to each site by the investigator. Backup copies of the computer disks containing data generated in each session were made before the investigator left the site.

Pretesting

Validation studies have shown that a learning effect takes place between the first and second attempts at the test battery but not significantly thereafter (Baker *et al.*, 1985). Therefore, a training session, or pretest, was scheduled for

each firm during the winter or early spring of 1991. The investigator transported 3 to 6 computers, depending on the number of subjects participating, to a facility, usually the firm's office, identified by the safety officer of each firm. The investigator gave an oral presentation explaining the purpose and scope of the project. Each subject interested in participating was asked to complete a personal data form (Appendix B) and a medical history form (Appendix C), and to sign a consent form (Appendix D). The subjects were then run through the test battery just as they would later experience it when data were being collected.

Facilities

During the winter of 1990 - 1991, potential participants were grouped by location, convenient facilities for administering ChE and NES2 tests were identified, and a master schedule was drawn up. Seven facilities were chosen, including the conference rooms of 5 participating firms, a conference room at the University of Maryland, and a classroom at Montgomery Community College. The facilities were set up in accordance with CDC/NIOSH training received by the investigator. To minimize experimental error, every attempt was made to ensure uniformity of working space and of placement of computer keyboards and joysticks. The rooms were arranged as similarly as their different configurations would allow, with either two 8-foot tables or three 6-

foot tables. Divider boards were constructed out of 2-foot by 4-foot sheets of lauan held in place by 8-inch 2-by-4's with grooves cut into them to hold the boards. Dividers were placed along the side of each computer to separate it from the others. In cases where the computers did not directly face a wall, another divider board was added in back of the computer such that each computer was contained in a cubicle of similar size.

Signs requesting quiet were posted on all entrance doors to the conference rooms. The signs also stated whether the next subject could enter or should wait outside so that subjects already performing the tests would not be disrupted. As there were 6 computers, 6 subjects could take the tests at any one time. In most cases, the maximum number of subjects scheduled on any one date was 12, ensuring that no subject would normally have to wait more than 40 minutes to begin the tests. In fact, even when as many as 17 subjects were scheduled on one date, due to the difference in what time subjects finished their work day most subjects did not have any waiting period, and the average waiting period for the others was about 5 minutes.

Supplemental information

Data on age, education level, and preexisting medical conditions were determined from the aforementioned personal data and medical history forms completed by all subjects during the pretest session. Sex and race were

determined by observation. On the advice of the IRB, sensitive questions were not asked until the end of the study in order to minimize the possibility of subjects refusing to participate. Additionally, the IRB felt that the subjects would be more likely to answer the questions truthfully at the end of the study.

During the blood sample collection for the fourth test session, an interview was conducted with each exposed subject to determine which pesticides the subject had used during the 1991 season and how frequently each had been applied by the subject. Immediately following the last NES2 test session, all subjects were asked to fill out a questionnaire (Appendix E,F) to investigate their level of nonoccupational exercise; frequency of use of alcohol, caffeine, and other drugs; and use of all pesticides in previous years. In addition, exposed subjects were questioned about the amount of protective clothing usually worn while working with insecticides and about previous training in pesticide safety.

Exposure potential

Level of exposure potential was estimated subjectively by asking the applicators at mid-season and at the end of the season to recall any OP products they had applied and to indicate the approximate number of times they had applied the materials. Factors considered in assignment of a subject to an exposure potential group included: 1.) average number of applications of

ChE-inhibiting insecticides per week during use period; 2.) percentage of the season that ChE-inhibiting insecticides were used; 3.) LD₅₀ values of ChE-inhibiting insecticides used; 4.) formulation of products used (emulsifiable concentrates tend to have a higher concentration of active ingredient and are more easily absorbed than granular materials); 5.) involvement in mixing and loading operations as well as application of ChE-inhibiting insecticides (mixing and loading operations generally provide more opportunity for exposure to concentrate pesticides than application activities), and 6.) amount of personal protective equipment usually worn by the subject during mixing/loading operations (if any) and during application.

Statistical Analyses

Analysis of variance was used to look at differences in neurobehavioral performance and in ChE outcome between the exposed group and controls on each of the five test dates. Subsequently, a more sensitive test, analysis of variance for repeated measures, was used to further test the hypothesis of an association between performance on neurobehavioral tests and OP exposure. Subjects were grouped by exposure potential level (high, medium, low, or control) and responses on NES2 tests, as well as percent change in ChE, were compared over time.

The planned contrasts were as follows:

(H vs. M vs. L) by session

(C vs. H + M + L) by session

(C vs. H) by session

(C vs. M) by session

(C vs. L) by session

where H, M, L, and C are high, medium, and low exposure potential and controls, respectively. Specifics of the analyses are discussed more fully in the Results section.

Results

Characteristics of the study population

Data developed from information taken during pretest sessions

All participants completed two questionnaires during the pretest session (Appendix B,C). Data described below were compiled from these forms and from observations made by the investigator.

The initial study population consisted of 52 exposed subjects and 39 controls from 17 firms. The number of participants from each firm ranged from 1 to 14. Although a high drop-out rate had been anticipated by the investigator, in fact, only 9 exposed and 2 control subjects dropped out. Several participants missed one or more sessions due to scheduling problems. Only those subjects completing the baseline and at least 2 other sessions were included in the analysis. Two subjects, originally assigned to the exposed group, were reassigned to the control group because they did not apply any OPs at any time during the season. Similarly, 1 control subject was reassigned to a spray job by his supervisor and so was reclassified into the exposed group. The final population thus consisted of 42 exposed and 38 controls (Table 3). Age distribution of the exposed group ranged from 22 to 43; the control group

ranged from 19 to 60 years. Both populations were comprised overwhelmingly of white males, and both displayed similar educational profiles.

No participants had ever been treated for symptoms of poisoning by an OP, and none had ever had a doctor diagnose them as suffering from any muscular disease, permanent muscle injury, or CNS pathology. Ten exposed subjects and 5 controls had sustained previous injuries to the head, but most injuries had occurred during childhood and all subjects had made complete recoveries. Two subjects in each group had arthritis, but all reported that it did not interfere with hand or finger sensation or movement. Two exposed subjects had peripheral neuropathies resulting from back injuries; in neither case did the neuropathy interfere with hand or finger sensation or movement. One control subject had been diagnosed with liver disease. One exposed subject had exhibited a seizure disorder in childhood but had not taken medication for the last 8 years and had not had a seizure in over 14 years.

Data developed from postseason questionnaire

All participants were asked to complete a questionnaire (Appendix E,F) at the end of the last test session. Three participants (1 control and 2 exposed subjects) missed the session and did not answer the request to respond to the questionnaire by mail. Therefore, results described below do not total 100%.

Control subjects had been employed in their current jobs an average of

6 years; 26% had worked with insecticides in previous jobs, for an average of 5 years. Exposed subjects had been employed in their current jobs an average of 4 years; 36% had worked with insecticides previously, averaging 8 years.

Subjects' use of insecticides at home during the test period did not differ between the control and exposed groups. Most participants used insecticides at home only about once a month (53.6% of exposed, 56.8% of controls). A large proportion (41.5% of exposed, 40.5% of controls) did not use them at all during the test period.

Twenty-six of the exposed subjects were certified applicators, and another 9 were registered technicians. (Certified applicators receive in-depth training and must pass an exam on various aspects of safe use of pesticides; registered technicians receive more basic training and do not take an exam.) The majority of exposed subjects (69.2%) had received at least 1 full day of training before they began working with insecticides. Another 20.5% had received some safety training, but less than 1 full day. During the entire time the exposed subjects had been working with insecticides, all had received some amount of safety training: 84.6% had received an average of more than 1 full day per year, 12.8% averaged 1 day per year, and 2.5% averaged less than 1 day per year.

The number of exposed subjects (19.0%) reporting frequent participation in vigorous non-work related exercise was somewhat higher than the control group (5.3%). Another 57.1% of exposed subjects and 68.4% of control subjects reported occasional participation in vigorous exercise. Those reporting

no vigorous exercise constituted 16.7% of the exposed population and 23.7% of the controls.

No subjects reported any use of illegal drugs other than marijuana during the ten-month study period (Table 4). Use of tobacco, alcohol, and caffeine was similar between the two groups. Two control subjects were daily users of blood pressure medications that can slow reaction time; medications regularly taken by other subjects are not associated with neurobehavioral effects (Table 5).

None of the non-cholinesterase-inhibiting pesticides used occupationally by any subjects (Table 6) have been associated with adverse neurobehavioral effects. During the 15 years prior to the study, exposed subjects had used a wider variety of pesticides at work than did controls (Tables 7,8). Only approximately one third of the control subjects had used at least one OP or carbamate insecticide in their work at some time during the previous 15 years.

Exposure potential classification

Contributing factors

Exposure potential classification was based primarily on which ChE-inhibiting insecticides the subject used and how often the subject applied them. Additional factors that could affect final classification included: the formulation

of products applied; whether the subject was exposed to concentrates during mixing and loading activities or only to already diluted sprays during application; and the level of protective gear worn during mixing/loading and during application.

More than three-quarters of the applicators reported using carbaryl and chlorpyrifos, with acephate also being used by a large proportion (Table 9). Both the active ingredient and the formulation were considered in assigning an applicator to an exposure group. Due to the lack of a better predictor of neuropsychological-potentiating effect, the oral LD₅₀ of the insecticides was used. Thus, use of materials with lower LD₅₀ values would tend to move a subject into the high exposure group. However, the product formulation was also taken into account. Granular formulations generally have less potential for absorption and would thus tend to move a subject into the low exposure group compared to use of an emulsifiable concentrate of the same active ingredient. Carbamates are reversible inhibitors and as such they would seem to be less capable of causing behavioral effects through the mechanisms investigated in this experiment. Therefore, use of carbamate insecticides had a neutral effect on classification of exposure level.

Most subjects wore appropriate amounts of protective gear during mixing/loading operations and during application of insecticides (Tables 10,11). During the 1991 spray season, 38 exposed subjects (90.5%) were involved in mixing operations on the job.

Classification

Five subjects applied OPs between 1 and 5 times during the entire season (approximately 20 weeks). Due to the infrequency of OP use, these subjects were classified as low exposure potential.

The remaining 37 subjects applied at least 1 OP at least once a week over at least 50% of the season. Eighteen of the 37 subjects used dicotophos, isofenphos, oxydemeton-methyl, and/or liquid formulations of diazinon and chlorpyrifos at least 4 times per week over at least 70% of the season. Due to the low LD₅₀ values of these insecticides and the frequency of use, these 18 applicators were classified as high exposure potential. Most also used other OPs and carbamates at some time during the season.

Twelve subjects used only malathion, acephate, carbaryl, and/or granular formulations of diazinon, isofenphos, and bendiocarb. Due to the relatively high LD₅₀ values of malathion and acephate, the fact that bendiocarb and carbaryl are carbamates, and the low absorbability of the granular formulations, these applicators were classified as medium exposure potential.

One subject applied liquid diazinon 3 times a week for 70% of the season, but was not involved in mixing and loading operations and wore additional protective gear (rubber gloves and rubber boots) during application. Another subject applied liquid chlorpyrifos 1 to 2 times a week throughout the season, and wore goggles as extra protection during mixing and loading

operations. A third subject applied liquid chlorpyrifos for 10% of the season. These 3 subjects were classified as medium exposure potential.

Four subjects applied liquid diazinon and/or liquid chlorpyrifos daily for at least 50% of the season; 2 of these subjects wore leather boots rather than rubber boots. These subjects were classified as high exposure potential.

Thus, this classification scheme resulted in 5 applicators in the low exposure potential group, 15 in the medium exposure potential group, and 22 in the high exposure potential group.

ChE effects

For each session subsequent to the first, the percent change in ChE level was calculated as

$$100 \left(\frac{\text{ChE}_s - \text{ChE}_b}{\text{ChE}_b} \right)$$

where ChE_b = baseline ChE (session 1)

ChE_s = ChE at session 2, 3, 4, or 5

Mean plasma ChE for both the medium and high exposure potential groups remained within 5% of baseline values throughout the study period; means for the low exposure potential group decreased during the season and began to return to baseline at the postseason session; control means were intermediate (Figure 3). Mean RBC ChE in all exposure potential groups

(including controls) tended to drop throughout the study period (Figure 4). Standard deviations ranged from 1.34 to 13.68 (Table 12). Comparisons of the pattern of change of ChE for individual subjects between session 2 and baseline (Figures 5 - 10) are representative of similar comparisons for sessions 3 through 5.

During the course of the study, 10 subjects exhibited a drop in either RBC ChE or plasma ChE of 25 percent or more compared against their own baseline values (Table 13). Four subjects exhibited this level of depression in RBC ChE, 5 subjects in plasma ChE, and 1 subject in both. The largest decrease in plasma ChE was a decrease of 46.8% of baseline; the corresponding figure for RBC ChE was 39.0%. Probable causes were determined by telephone interviews with the subjects at the time the decreases were discovered. Four cases were probably due to overexposure at work. One control subject was exposed at home through treating his entire yard with the OP diazinon while not wearing any protective gear. He then went to work and climbed a tree still wet with an OP. Two other control subjects, co-workers at a park facility, treated the apartment building of one of the subjects for cockroaches. The name of the pesticide they used could not be determined, but the active ingredient was probably chlorpyrifos or diazinon. Two subjects were hospitalized for reasons unrelated to OP exposure shortly before testing; either their condition or their treatment may have affected their ChE activity levels. The reason for ChE decrease in 1 control case could not be determined. It is

possible that the subject applied an OP either at home or at work but was afraid to admit it to the investigator.

Effects of exposure on NES2 test performance

The NES2 data were stored on disks and the SAS System (version 6.03) was used to perform statistical analyses on a main frame computer. Scores used as dependent variables for each of the NES2 tests are identified in Table 14.

Subjects who missed a scheduled test session were rescheduled for another session. However, a few subjects could not be rescheduled. Two exposed subjects each missed 1 entire session (ChE testing and NES2 tests) and 2 missed 2 sessions. One exposed subject missed 1 entire session and 1 ChE test on another occasion. One exposed subject missed only the ChE testing on 1 occasion. Four control subjects each missed 1 entire session. Analysis of variance for repeated measures requires that all subjects have data at each time. A repeated measures analysis was run on subjects for whom data existed for each test at each session in order to examine the lack of independence. The covariance technique of Steel and Torrie (1960) was used to estimate missing data for these subjects, and the repeated measures analysis was adjusted to account for the degree of correlation and the missing values.

Residuals were examined for assumptions of normality and homogeneity of variance; data did not need transformation. Because missing data were

estimated, and repeated measures are not independent, the degrees of freedom were adjusted to provide valid estimates of the probabilities for the analysis of variance.

Performance as a function of exposure vs. control group

An analysis of variance (ANOVA) was applied to the data to test for differences in NES2 performance between control and exposed groups in each of the five test sessions. Statistical significance was found in 7 out of the 110 comparisons ([20 test scores + 2 ChE values] x 5 sessions) at the 0.05 level. The affected tests were 1.) reaction time on the continuous performance test (at each of the 5 sessions) and 2.) reaction time on the symbol-digit substitution test (sessions 1 and 3 only). In each case, control subjects performed more poorly (slower) than the exposed group.

Performance as a function of exposure potential group over time

Univariate repeated measures analysis was used to compare the exposure potential groups on the basis of their change in ChE levels and their performance on NES2 tests over time.

F statistics were calculated for each of 20 NES2 test scores and RBC and plasma ChE levels. The comparisons of interest are presented in Table 15.

Results of the planned contrasts described in the methods section were largely nonsignificant, but F statistics on 5 of the comparisons were statistically significant at the 0.05 level (Table 16). In 3 of the NES tests, the comparison of H, M, and L across time was found to be significantly different, thus a total of 4 comparisons were performed for each of those tests. On the remaining tests, the initial comparison was not significantly different, therefore 2 comparisons were made for each of those tests. A total of 50 comparisons were made. Neither the continuous performance test nor the symbol-digit substitution test, found to differ statistically in the test described above, was among those found to be statistically significant by this method. The direction and patterns of significance were not consistent.

Performance as a function of percent change in ChE

Linear regressions were calculated, with percent change in ChE (from baseline) as the independent variable and the NES2 scores as the dependent variables. A regression line was developed for each test score on each of the 4 sessions subsequent to baseline. F statistics for both plasma ChE change and RBC ChE change on 3 of the 80 comparisons (20 test scores for each of 4 sessions) were statistically significant at the 0.05 level (Table 17). Direction of significance showed poorer performance by subjects in the exposed group compared to controls in 5 out of the 6 statistically significant cases. The

difference in number of errors on the fingertapping test was statistically significant for both plasma and RBC ChE in session 3. While total number of taps (fingertapping test) was significantly different in both plasma and RBC ChE regressions, the affected sessions differed and the trials were not equivalent. The first and second trials of this test measure the number of taps achieved with the preferred hand (the right hand for most people), the third trial measures performance with the nonpreferred hand, and the fourth trial measures performance of alternating hands.

A correlation analysis was run comparing percent ChE change in subjects with a 25 percent or greater depression with change in NES performance for that session. Due to the small number of subjects exhibiting such a drop, plasma and RBC ChE values were combined for this analysis; *i.e.*, if the subject's plasma ChE level decreased from baseline by 30 percent but his RBC ChE level decreased by only 15 percent of baseline, the plasma ChE change for that subject was entered into the analysis. If a second subject exhibited a 15 percent decrease in plasma ChE and a 30 percent decrease in RBC ChE, the RBC ChE value was entered. Sessions were also combined in this analysis; only data for the session in which the subject's ChE decreased level was detected were entered. This method resulted in 6 plasma ChE values and 5 RBC ChE values from a total of 10 subjects (1 subject exhibited a decrease of the required magnitude in both plasma and RBC ChE). Because only eleven cases were evaluated, the decision was made to accept a correlation coefficient of 0.60

(absolute value) or greater as an indication of a possible correlation. A correlation coefficient of this magnitude occurred only for the simple reaction time test; number of errors on the fingertapping test showed a correlation coefficient of -0.59 (Table 18). Decreased ChE level exhibited a positive relationship with reaction time and a negative relationship with the number of errors on the fingertapping test.

Sensitivity of analyses

In this study, the ANOVA at a single session generally had a 90 percent chance of detecting differences of approximately 4 to 15 percent on tests measuring perceptual-motor performance and memory, and a 15 to 23 percent difference in tests measuring affect (Table 19). These estimates would be conservative with respect to the repeated measures analyses, as this design is more sensitive than the ANOVA performed on data for each session.

Table 3.--Characteristics of the study group.

	<u>Exposed</u>	<u>Controls</u>
Number of subjects	42	38
Age, mean years	29.2	34.7
Age distribution (no./%)		
<25	12 (28.6)	8 (21.0)
26 - 34	24 (57.1)	15 (39.5)
35 - 44	6 (14.3)	10 (26.3)
>45	0 (0)	5 (13.2)
Sex (no./% males)	38 (90.5)	32 (84.2)
Ethnicity (no./%)		
White	38 (90.5)	32 (84.2)
Black	2 (4.8)	5 (13.2)
Hispanic	2 (4.8)	1 (2.6)
Education, mean years	14	14
Education distribution (no./%)		
< high school	0 (0)	3 (7.9)
high school	10 (23.8)	11 (28.9)
some college	16 (38.1)	10 (26.3)
Bachelor's degree	15 (35.7)	11 (28.9)
graduate training	1 (2.4)	3 (7.9)

Table 4.--Subjects' use of caffeine, tobacco, alcohol, and recreational drugs^{1,2}.

SUBSTANCE	DAILY		1-2X/WEEK		1-2X/MONTH		NEVER	
	EXP	CTL	EXP	CTL	EXP	CTL	EXP	CTL
Caffeine	88.0	84.2	4.8	7.9	2.4	5.2	0	2.6
Tobacco	31.0	39.4	2.4	5.3	4.8	5.2	57.1	50.0
Alcohol	11.9	10.5	42.8	44.7	28.6	31.6	11.9	13.2
Marijuana	0	0	4.8	0	9.5	5.3	78.6	92.1
Cocaine	0	0	0	0	0	0	95.2	97.4
Other illegal drugs	0	0	0	0	0	0	95.2	97.4

¹Expressed as percent of subjects in each group; EXP = exposed (n = 42); CTL = control (n = 38).

²Results do not total 100% because 2 exposed and 1 control subject failed to return the questionnaire.

Table 5.--Subjects' regular use of prescription medications during the test period.

<u>Brand name</u>	<u>Generic name</u>	<u>Type of drug</u>	<u>No. of sub-jects¹</u>	<u>Frequency of use</u>
Brevicon	norethindrone + ethinyl estradiol	oral contraceptive	1CTL	daily
Flexeril	cyclobenzaprene hydrochloride	skeletal muscle relaxant	2EXP	2x/month
Inderal	propranolol hydrochloride	β -adrenergic receptor blocker	1CTL	daily
Ovral	norgestrel + ethinyl estradiol	oral contraceptive	1EXP	daily
Motrin	ibuprofen	nonsteroidal anti-inflammatory agent; prostaglandin inhibitor	1EXP	daily
Naprosyn	naproxen	nonsteroidal anti-inflammatory agent; prostaglandin inhibitor	1EXP	2x/month
Seldane	terfenadine	histamine H ₁ receptor antagonist	1EXP	daily
Synthroid	levothyroxine sodium	thyroid hormone replacement	1EXP	daily
Tenormin	atenolol	β -adrenergic receptor blocker	1CTL	daily

¹EXP = exposed; CTL = control

Table 6.--Number of subjects using non-cholinesterase-inhibiting pesticides at work during the 1991 season.

	<u>Exposed (n = 42)</u>	<u>Control (n = 38)</u>
<u>Insecticides/miticides</u>		
<u>Biologicals</u>		
Bt	11	2
pyrethrins	9	0
rotenone	2	0
<u>Inorganic compounds</u>		
horticultural oil	1	0
insecticidal soap	13	1
<u>Insect growth regulators</u>		
diflubenzuron	9	5
<u>Organochlorines</u>		
dicofol	20	2
dienochlor	8	2
lindane	3	3
methoxychlor	8	2
oxythioquinox	8	0
<u>Pyrethroids</u>		
cyfluthrin	14	5
fluvalinate	13	3
<u>Fungicides</u>		
chlorothalonil	15	1
cycloheximide	1	0
ferbam	6	0
iprodione	7	0
metalaxyl	6	2
triadimefon	11	0
<u>Herbicides</u>		
2,4-D	1	0
DCPA	2	0
dicamba	17	3
glyphosate	24	19
isoxaben	0	1
mecoprop	1	0
metolachlor	0	1
oryzalin	0	1
picloram	0	1
triclopyr	2	0

Table 7.--Number of exposed subjects (n = 42) using pesticides at work in previous years.

	<u>90</u>	<u>89</u>	<u>88</u>	<u>87</u>	<u>86</u>	<u>85</u>	<u>84</u>	<u>83</u>	<u>82</u>	<u>81</u>	<u>80</u>	<u>79</u>	<u>78</u>	<u>77</u>	<u>76</u>
<u>Insecticides/miticides</u>															
<u>Organophosphates</u>															
acephate	10	10	9	7	5	4	4	4	3	2	2	2	2	1	1
chlorpyrifos	12	14	12	10	7	4	4	3	3	2	2	1	1	0	0
diazinon	13	13	11	6	6	6	6	4	4	3	3	2	2	1	1
dicrotophos	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0
dimethoate	2	2	3	3	2	2	2	1	1	0	0	0	0	0	0
isazophos	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0
isofenphos	2	5	5	5	3	2	2	1	0	0	0	0	0	0	0
malathion	6	4	3	4	3	3	3	3	3	2	2	2	2	1	1
oxydemeton-methyl	1	2	2	2	2	1	2	0	0	0	0	0	0	0	0
trichlorfon	1	1	1	0	0	0	0	0	1	1	1	0	0	0	0
<u>Carbamates</u>															
bendiocarb	3	2	2	1	0	0	0	0	0	0	0	0	0	0	0
carbaryl	13	11	7	7	5	3	3	3	3	2	2	2	2	1	1
<u>Biological compounds</u>															
Bt	7	4	2	2	1	0	0	0	0	0	0	0	0	0	0
pyrethrins	4	3	3	3	4	2	1	1	1	0	0	0	0	0	0
rotenone	2	2	1	1	1	1	1	2	2	0	0	0	0	0	0
<u>Inorganic compounds</u>															
insecticidal soap	9	5	2	1	1	0	0	0	0	0	0	0	0	0	0
<u>Insect growth regulators</u>															
diflubenzuron	7	5	4	2	0	0	0	0	0	0	0	0	0	0	0
<u>Organochlorines</u>															
chlordane	0	0	0	1	1	1	1	1	0	0	0	0	0	0	0
dicofol	6	6	8	8	5	3	3	3	2	1	1	1	1	1	1
dienochlor	4	4	1	1	2	1	0	0	0	0	0	0	0	0	0
lindane	3	3	2	1	1	1	1	2	1	1	1	1	1	1	1
methoxychlor	4	4	3	3	4	3	3	3	2	1	1	1	1	1	1
oxythioquinox	2	4	2	2	1	0	0	0	0	0	0	0	0	0	0
<u>Pyrethroids</u>															
cyfluthrin	6	2	0	0	0	0	0	0	0	0	0	0	0	0	0
fluvalinate	7	5	2	0	0	0	0	0	0	0	0	0	0	0	0

Table 7. (cont'd.)

	<u>90</u>	<u>89</u>	<u>88</u>	<u>87</u>	<u>86</u>	<u>85</u>	<u>84</u>	<u>83</u>	<u>82</u>	<u>81</u>	<u>80</u>	<u>79</u>	<u>78</u>	<u>77</u>	<u>76</u>
<u>Fumigants</u>															
aluminum															
phosphide	0	0	0	0	1	1	1	1	1	0	0	0	0	0	0
methyl bromide	0	0	0	0	1	1	1	1	1	0	0	0	0	0	0
<u>Fungicides</u>															
anilazine	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
benomyl	3	4	3	3	2	1	1	1	0	0	0	0	0	0	0
chlorothalonil	1	3	2	2	1	1	1	0	0	0	0	0	0	0	0
cycloheximide	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0
fenarimol	0	1	1	0	0	0	0	0	0	0	0	0	0	0	0
ferbam	3	3	4	1	0	0	0	0	1	1	1	0	0	0	0
iprodione	5	4	1	1	0	0	0	0	0	0	0	0	0	0	0
maneb	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
metalaxyl	2	0	0	0	1	1	0	0	0	0	0	0	0	0	0
triadimefon	6	4	4	2	1	0	0	0	0	0	0	0	0	0	0
<u>Herbicides</u>															
DCPA	0	0	0	0	1	1	1	1	1	0	0	0	0	0	0
dicamba	9	10	7	7	5	5	5	3	2	2	2	0	0	0	0
fenoxaprop-ethyl	0	1	1	0	0	0	0	0	0	0	0	0	0	0	0
glyphosate	11	11	7	7	4	2	2	0	0	0	0	0	0	0	0
triclopyr	2	2	2	1	1	0	0	0	0	0	0	0	0	0	0

Table 8.--Number of control subjects (n = 38) using pesticides at work in previous years.

	<u>90</u>	<u>89</u>	<u>88</u>	<u>87</u>	<u>86</u>	<u>85</u>	<u>84</u>	<u>83</u>	<u>82</u>	<u>81</u>	<u>80</u>	<u>79</u>	<u>78</u>	<u>77</u>	<u>76</u>
<u>Insecticides/miticides</u>															
<u>Organophosphates</u>															
acephate	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1
azinphosmethyl	0	0	1	1	1	1	1	1	1	1	1	1	1	1	1
chlorpyrifos	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1
diazinon	1	3	2	2	1	1	1	1	1	1	1	1	1	1	1
dimethoate	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1
malathion	0	1	0	0	0	0	0	0	0	1	1	1	1	1	1
methyl parathion	0	0	1	1	1	1	1	1	1	1	1	1	1	1	1
parathion	0	0	1	1	1	1	1	1	1	1	1	1	1	1	1
<u>Carbamates</u>															
carbaryl	1	0	0	0	0	0	0	0	1	1	2	2	2	2	1
<u>Biological compounds</u>															
Bt	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0
pyrethrins	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0
<u>Inorganic compounds</u>															
insecticidal soap	1	1	0	1	0	0	0	0	0	0	0	0	0	0	0
<u>Organochlorines</u>															
dicofol	0	0	0	0	1	1	1	1	1	1	1	1	1	1	1
lindane	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1
methoxychlor	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1
<u>Fumigants</u>															
methyl bromide	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1
<u>Fungicides</u>															
benomyl	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0
ferbam	0	0	0	0	0	0	0	0	0	0	1	1	0	0	0
metalaxyl	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0
triadimefon	0	0	1	1	1	0	0	0	0	0	0	0	0	0	0
<u>Herbicides</u>															
DCPA	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0
dicamba	2	1	1	1	1	1	1	1	1	1	1	1	1	1	1
glyphosate	5	7	6	5	4	4	3	2	2	2	2	2	1	1	1
paraquat	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1
triclopyr	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Table 9.--Cholinesterase-inhibiting insecticides used by exposed subjects (n = 42) during the 1991 spray season.

<u>Insecticide</u>	<u>LD₅₀¹</u>	<u>Percent of Subjects</u>
<u>Organophosphates</u>		
chlorpyrifos (Dursban)	96 ²	76.9
acephate (Orthene)	945 ³	66.7
diazinon (Diazinon)	66 ²	48.7
isofenphos (Oftanol)	20 ³	30.8
dimethoate (Cygon)	215 ²	20.5
malathion (Malathion)	480 ²	20.5
dicrotophos (Bidrin)	17 ³	2.4
oxydemeton-methyl (Metasystox-R)	30 ³	2.4
trichlorfon (Dylox)	250 ³	2.4
isazophos (Triumph)	50 ³	2.4
<u>Carbamates</u>		
carbaryl (Sevin)	246 ²	79.5
bendiocarb (Turcam)	40 ²	12.8

¹Oral rat; mg technical ingredient/kg body weight

²EXTOXNET, Cornell University, Ithaca, NY, 1991

³Farm Chemicals Handbook, Meister Publishing Co., Willoughby, OH, 1992

Table 10.--Protective gear usually worn by applicators (n = 42) when mixing concentrates¹.

<u>Gear</u>	<u>Percent</u>
<u>Clothing</u> ²	
Disposable outfit: long pants, long sleeves	7.7
Company outfit: long pants, long sleeves	25.6
Company outfit: long pants, short sleeves	66.7
Own clothing: long pants, long sleeves	5.1
Own clothing: long pants, short sleeves	7.7
Own clothing: shorts, short sleeves	0
<u>Gloves</u> ³	
Rubber or neoprene	82.0
Cotton	5.1
No gloves	5.1
<u>Footgear</u> ⁴	
Rubber or neoprene boots	64.1
Leather boots or shoes	38.5
Canvas shoes	2.6
Sandals	0
<u>Other</u>	
Hat	23.1
Goggles	33.3
Respirator or gas mask	35.9

¹Two subjects could not be reached to answer the questionnaire.

²Six subjects gave two answers; one subject gave three answers.

³One subject did not answer this question; one subject gave two answers.

⁴Two subjects gave no answer to this question; five subjects gave two answers.

Table 11.--Protective gear usually worn by applicators (n = 42) when applying insecticides¹.

<u>Gear</u>	<u>Percent</u>
<u>Clothing</u> ²	
Disposable outfit: long pants, long sleeves	2.4
Company outfit: long pants, long sleeves	23.8
Company outfit: long pants, short sleeves	69.0
Own clothing: long pants, long sleeves	2.4
Own clothing: long pants, short sleeves	2.4
Own clothing: shorts, short sleeves	4.8
<u>Gloves</u> ³	
Rubber or neoprene	69.0
Cotton	2.4
No gloves	16.7
<u>Footgear</u> ⁴	
Rubber or neoprene boots	71.4
Leather boots or shoes	28.6
Canvas shoes	0
Sandals	0
<u>Other</u>	
Hat	31.0
Goggles	26.2
Respirator or gas mask	26.2

¹Two subjects could not be reached to answer the questionnaire.

²One subject did not answer this question; three subjects gave two answers; one subject gave three answers.

³Four subjects did not answer this question; one subject gave two answers.

⁴Two subjects gave no answer to this question; five subjects gave two answers.

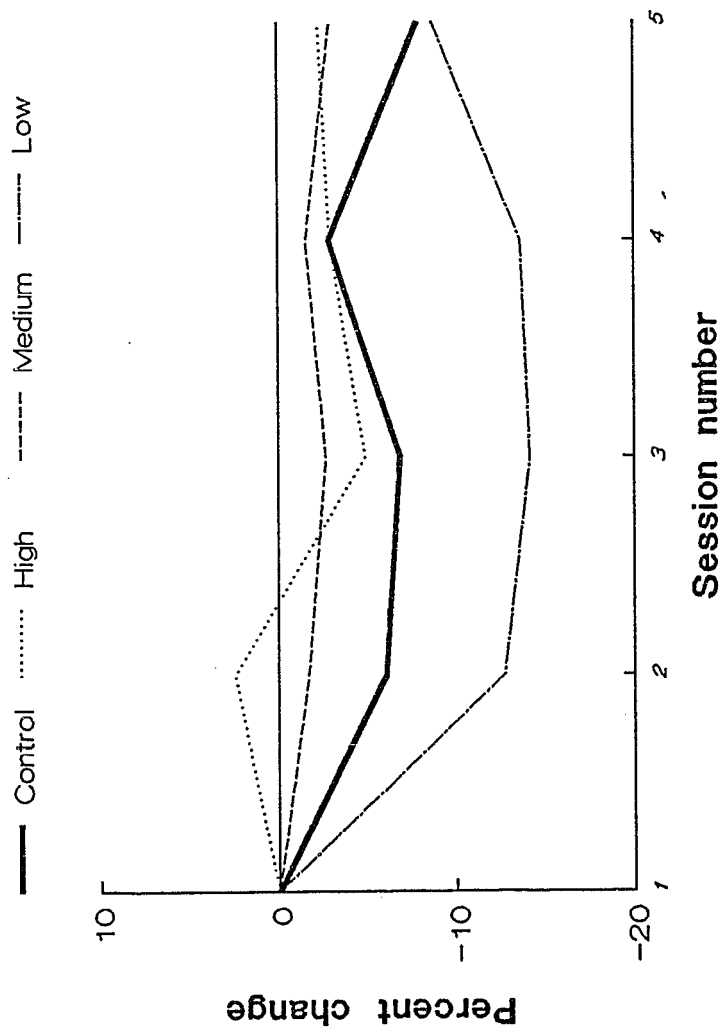


Figure 3.--Plasma ChE mean percent change from baseline.

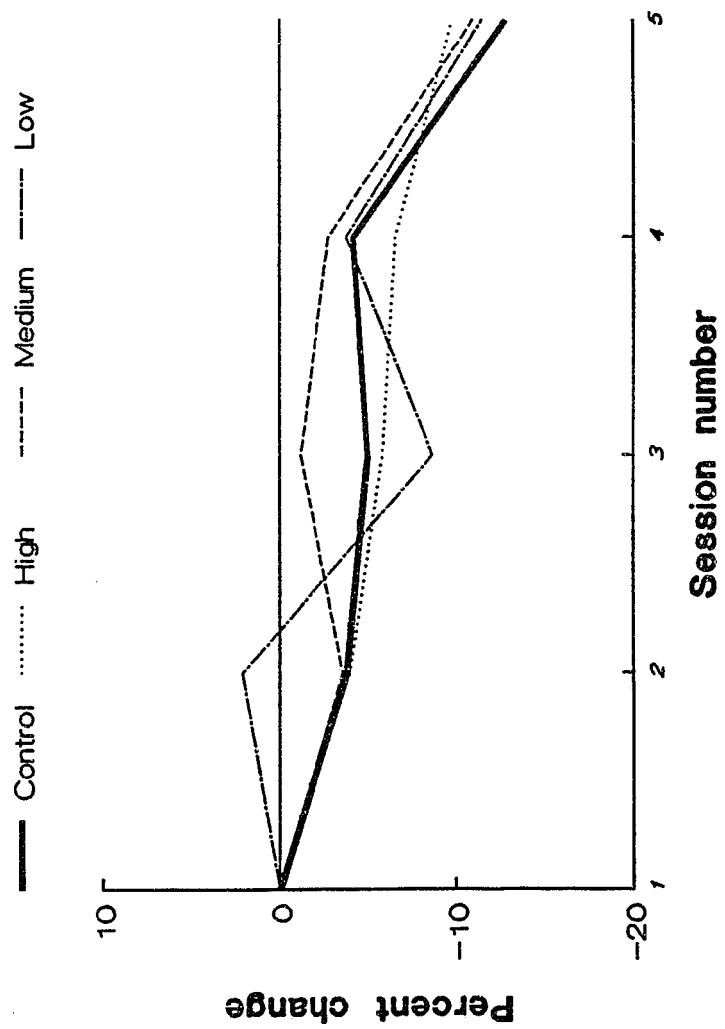


Figure 4.--RBC ChE mean percent change from baseline.

Table 12.--Means and standard deviations for ChE percent change from baseline¹.

Session No.	<u>High</u> RBC plasma		<u>Medium</u> RBC plasma		<u>Low</u> RBC plasma		<u>Control</u> RBC plasma	
2	-4.00 (5.88)	+2.49 (8.65)	-3.58 (6.89)	-1.69 (14.16)	+2.08 (13.68)	-12.66 (4.90)	-3.78 (7.55)	-6.06 (10.05)
3	-5.80 (7.86)	-4.89 (10.85)	-1.17 (1.34)	-2.67 (11.18)	-8.64 (11.35)	-14.10 (12.63)	-4.97 (8.17)	-6.84 (9.78)
4	-6.55 (9.97)	-2.92 (12.08)	-2.77 (7.75)	-1.55 (12.28)	-3.80 (4.01)	-13.58 (5.17)	-4.19 (8.61)	-2.86 (10.27)
5	-9.74 (9.58)	-2.28 (8.21)	-10.91 (8.52)	-2.95 (7.92)	-11.40 (7.40)	-8.70 (6.96)	-12.71 (7.92)	-7.88 (8.50)

¹Standard deviations in parentheses.

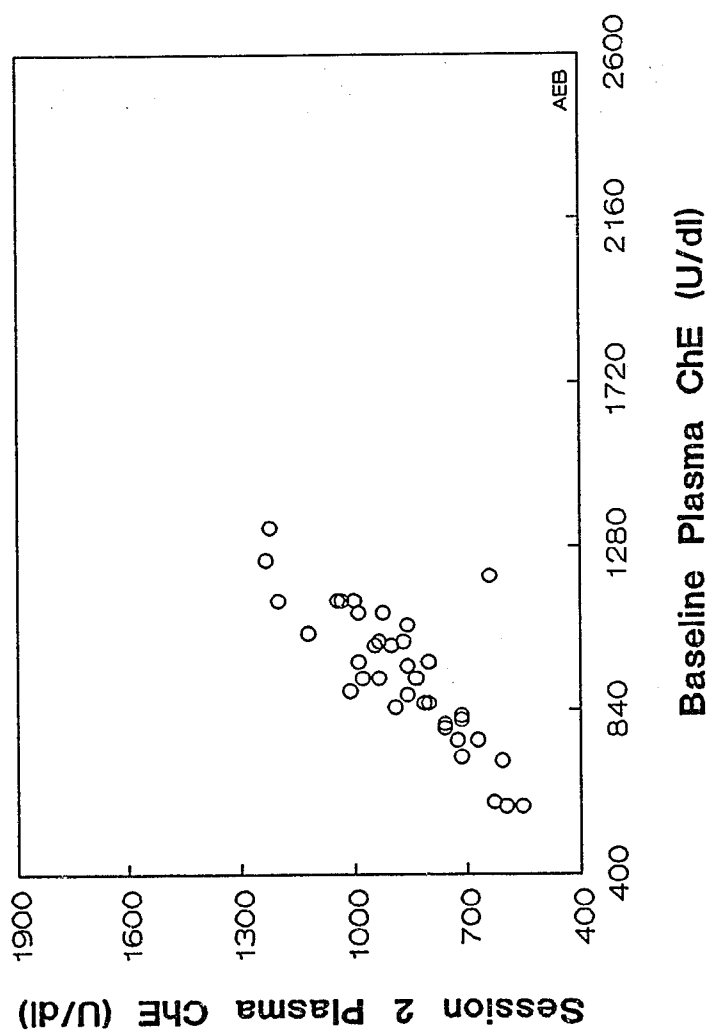


Figure 5.--Plasma ChE for session 1 (baseline) vs. session 2 -- Control subjects.

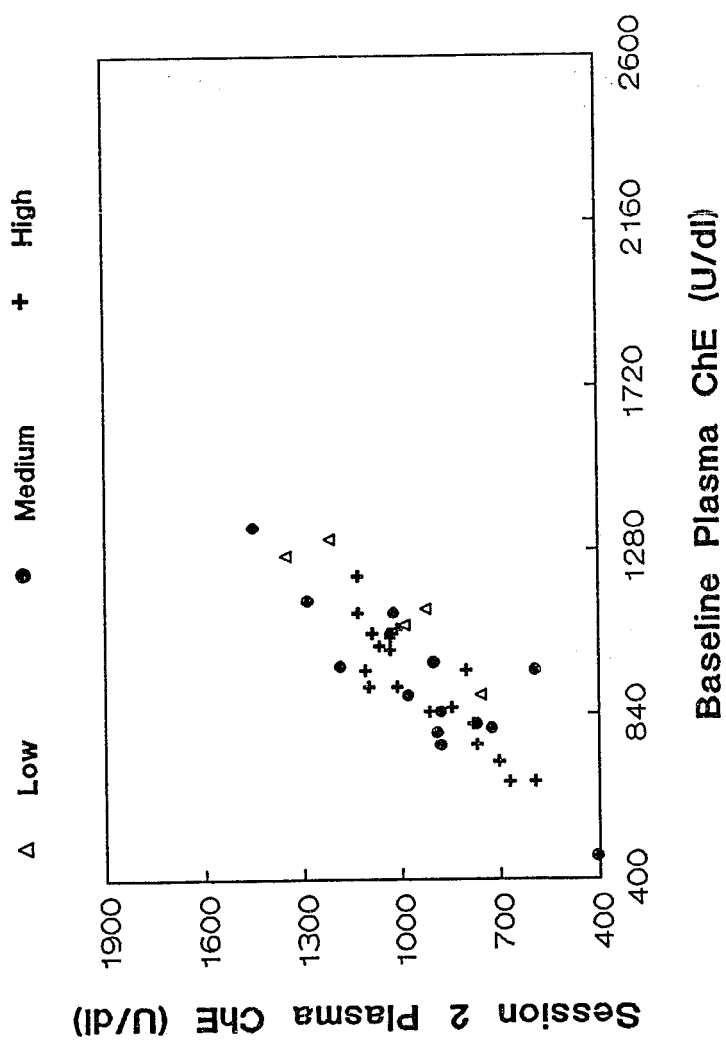


Figure 6.--Plasma ChE for session 1 (baseline) vs. session 2 -- Exposed subjects.

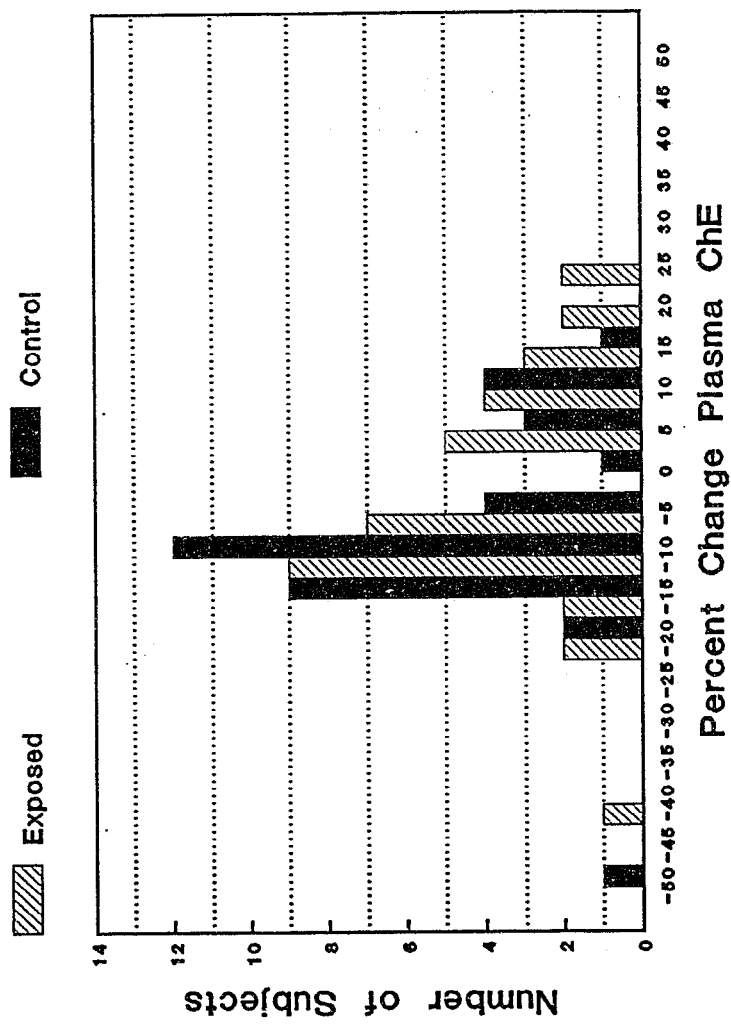


Figure 7.--Plasma ChE change: Comparison between baseline and session 2.

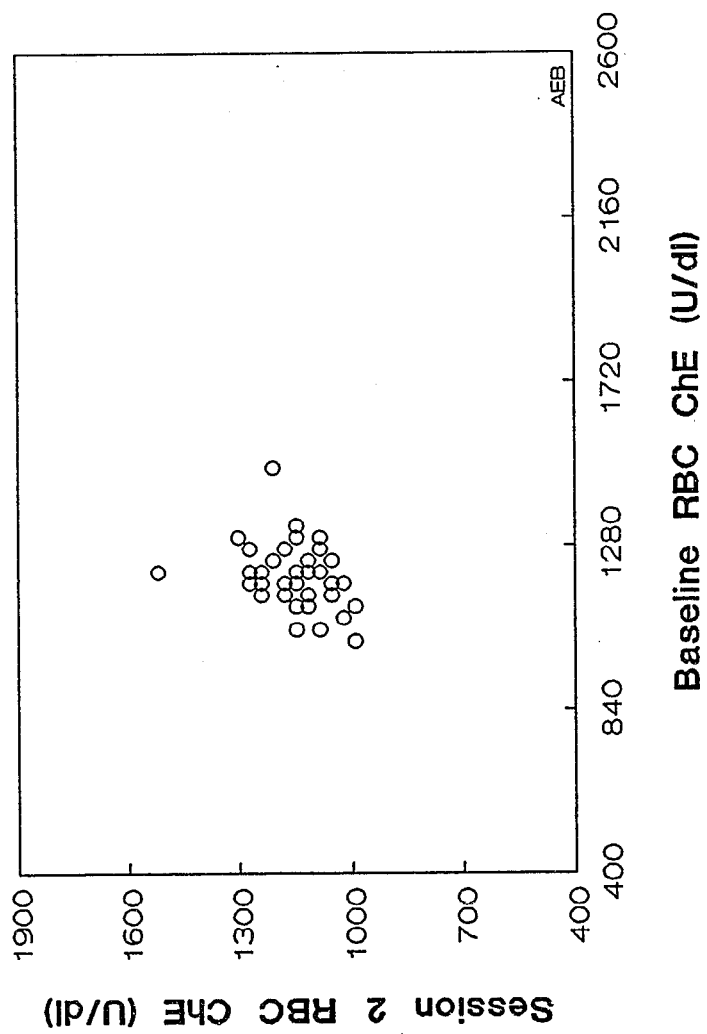


Figure 8.--RBC ChE for session 1 (baseline) vs. session 2 -- Control subjects.

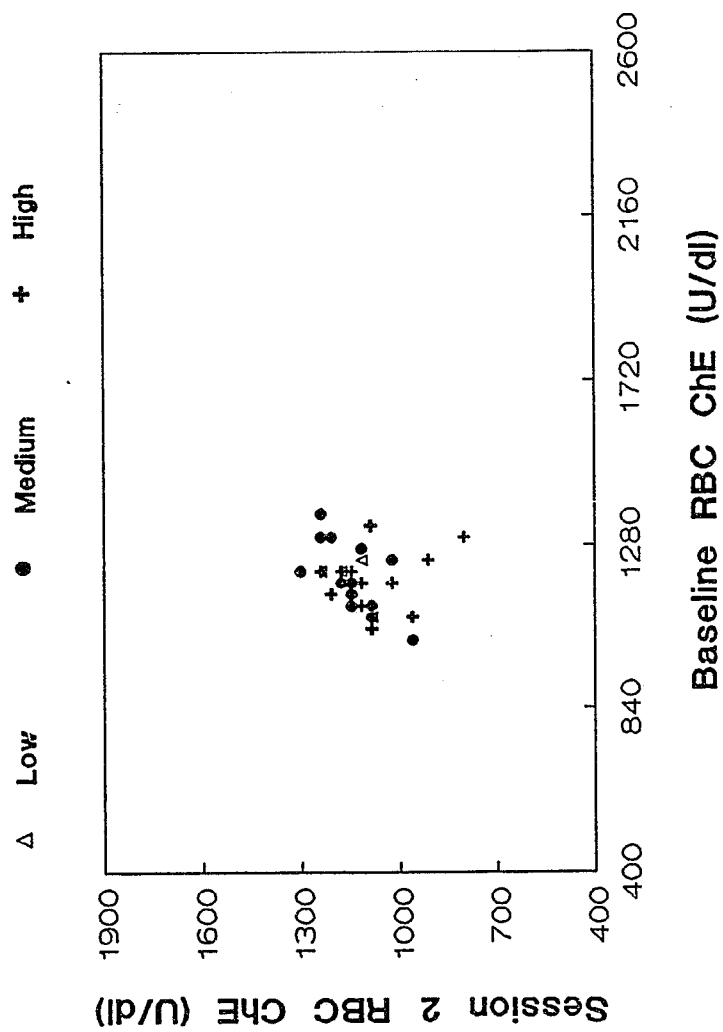


Figure 9.--RBC ChE for session 1 (baseline) vs. session 2 -- Exposed subjects.

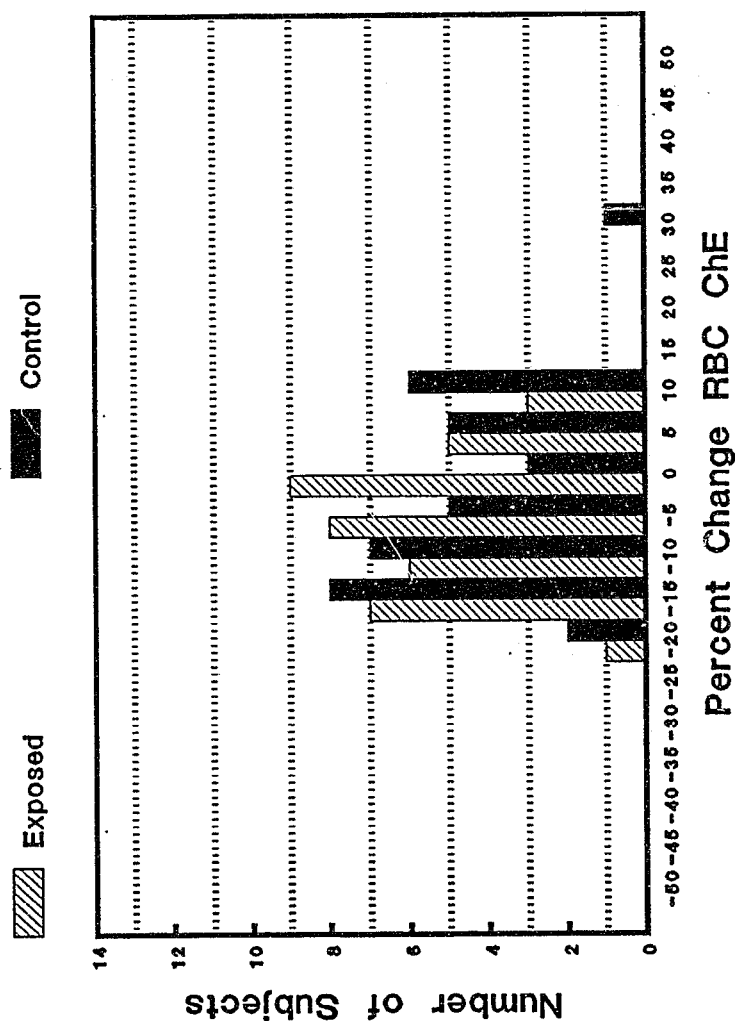


Figure 10.--RBC ChE change: Comparison between baseline and session 2.

Table 13.--Subjects with ChE decrease of 25% or more from baseline.

SUBJECT GROUP ¹	TEST SESSION	% ChE CHANGE (plasma)	%ChE CHANGE (RBC)	REASON
CTL	2	-46.8	-8.1	treated yard with diazinon without protective clothing; climbed treated tree at work
EXP _H	2	-37.9	-17.5	overexposure at work
EXP _L	3	-32.0	-23.1	applied chlorpyrifos at work without protective gear
CTL	3	-46.4	-28.2	unknown; hospitalized for colitis
EXP _H	3	-6.5	-28.5	overexposure at work
EXP _H	4	-18.6	-39.0	overexposure at work
CTL	5	-19.4	-31.0	overexposure at home
CTL	5	-26.2	-13.2	unknown
CTL	5	-5.6	-35.9	overexposure at home
CTL	5	-25.6	-7.9	unknown; hospitalized for knee surgery

¹CTL = control; EXP_H = high exposure potential group; EXP_L = low exposure potential group.

Table 14.--Dependent variables for NES2 tests.

<u>Test</u>	<u>Score</u>
Fingertapping	Total number of taps on each of 4 trials Total number of errors on trial 4
Continuous performance	Mean latency (response time) for all trials Number of false positives Number of omissions
Hand-eye coordination	Root mean square error Mean absolute error
Symbol-digit substitution	Mean latency (response time) for all trials Number incorrect on last trial
Simple reaction time	Mean latency (response time) for all trials
Digit span	Highest number correct in forward trials Highest number correct in backward trials
Mood scales	Mean score (scale of 1 - 5) for each of the 5 scales (anger, confusion, depression, fatigue, tension)

Table 15.--Repeated measures analysis: Comparisons of interest.

<u>Comparison</u>	<u>Question</u>
1. (H vs M vs L)*day	Does the exposed group differ between exposure potential groups (high, medium, and low) over time? (If yes, then perform comparisons 2 - 4; if no, perform comparison 5.)
2. (C vs H)*day	Does the control group differ from the high exposure potential group over time?
3. (C vs M)*day	Does the control group differ from the medium exposure potential group over time?
4. (C vs L)*day	Does the control group differ from the low exposure potential group over time?
5. (C vs H + M + L)*day	Does the control group differ from the exposed group over time?

Table 16.--Repeated measures analysis: Statistical significance at the 0.05 level.

<u>Test</u>	<u>Comparison</u>	<u>p Value</u>
Fingertapping number of errors	(H vs M vs L)*day	0.043
Mood scales: mean score anger	(H vs M vs L)*day (C vs M)*day	0.043 0.047
fatigue	(H vs M vs L)*day	0.016
Plasma ChE percent change	(C vs H + M + L)*day	0.039

Table 17.--Statistical significance of linear regressions with % change in ChE (from baseline value) as the independent variable and NES2 performance score as the dependent variable.

<u>NES2 test score</u>	<u>p value</u>	<u>Direction¹</u>
<u>Plasma ChE</u>		
Fingertapping, number of errors, session 3	0.008	+
Fingertapping, total number of taps, trial 1 (of 4), session 5	0.045	+
Digit span, number correct backwards, session 5	0.049	+
<u>RBC ChE</u>		
Continuous performance test, number of false positives, session 2	0.007	--
Fingertapping, number of errors, session 3	0.042	+
Fingertapping, total number of taps, trial 4 (of 4), session 4	0.008	+

¹+ indicates predicted performance on NES2 test improved as amount of ChE increased relative to baseline;

-- indicates performance on NES2 test worsened as amount of ChE increased relative to baseline.

Table 18.--Correlation coefficients for NES statistics and ChE percent depression in subjects (n = 11) with 25% or greater depression of either plasma or RBC ChE¹.

<u>NES statistic</u>	<u>Correlation coefficient</u>
Fingertapping	
number of taps, trial 1	+0.21
number of taps, trial 2	+0.16
number of taps, trial 3	-0.07
number of taps, trial 4	-0.12
number of errors	-0.59
Continuous performance	
reaction time	+0.07
Hand-eye coordination	
root mean square error	-0.09
mean absolute error	-0.26
Symbol-digit substitution	
reaction time	-0.05
Simple reaction time	
reaction time	+0.60
Digit span	
longest span forward	-0.30
longest span backward	+0.25
Mood scales: mean score	
anger	-0.02
confusion	-0.18
depression	+0.33
fatigue	-0.10
tension	+0.22

¹Correlation statistics were calculated for the session in which the subject's ChE level (either plasma or RBC) exhibited a change from baseline of 25% or greater. (Significant at the .05 level when $|r| > 0.60$)

Table 19.--Sensitivity of analyses¹.

<u>NES statistic</u>	<u>Average score</u>	<u>Sensitivity (%)</u>
Fingertapping		
number of taps, trial 1	177.7 taps	5.4
number of taps, trial 2	173.0 taps	5.6
number of taps, trial 3	160.3 taps	6.8
number of taps, trial 4	226.5 taps	9.2
number of errors	21.8 errors	12.6
Continuous performance		
reaction time	369.8 msec	3.8
Hand-eye coordination		
root mean square error	8.1	14.3
mean absolute error	6.3	14.4
Symbol-digit substitution		
reaction time	247.6 msec	9.4
Simple reaction time		
reaction time	285.0 msec	8.7
Digit span		
longest span forward	7.0 numbers	9.6
longest span backward	6.1 numbers	11.8
Mood scales: mean score (scale of 1 - 5)		
anger	1.9	22.6
confusion	2.2	17.8
depression	2.0	20.0
fatigue	2.8	15.7
tension	2.5	17.7

¹Computed for n = 40, $\alpha = .05$, $\beta = .90$.

Discussion and Conclusions

A sparse and scattered pattern of statistical significance was shown across the various analyses used in this investigation, and the direction of significance was often not consistent with impairment of neurobehavioral function due to OP exposure.

ChE effects

Plasma and RBC ChE levels did not differ statistically during the season between the exposed and control groups. This is indicative of the relatively low OP exposure that was a design feature of this study.

Individual variation appeared to be well within the range reported by other researchers. The large standard deviations around the means of percent change in ChE in this study reflect the variability of this measure throughout the season. Both intra- and interindividual variation in ChE levels are normally relatively large. In a study of 193 subjects with no known exposure to ChE inhibitors, Brock and Brock (1990) found an average intraindividual variation over a period of 8 months in plasma ChE of 14%, with a range of 3% to 42%. Bowers *et al.* (1964) found that even when dose was carefully controlled by administering an OP to volunteers through a microsyringe, neither the final level of ChE depression (measured 24 hours postinjection) nor the rate of

depression could be predicted from the dose.

RBC ChE levels in all groups declined throughout the study. A similar seasonal decrease in ChE levels was detected in a study by Yeary (1984) in which mean levels of RBC ChE of adults not occupationally exposed to OPs continued to decline into November (data presented covered January and March through November). Plasma ChE values in that study showed a steady decline from March through July but began to return toward the baseline in August, although they did not reach the January level. In the current study, plasma ChE values remained within 8 percent (7 percent if those with an acute overexposure incident are deleted) of the baseline values for all groups except those in the low exposure potential. Due to the small number of applicators in the low exposure potential group, the pattern of ChE depression in this group should be treated skeptically.

True exposure levels, as indicated by ChE depression, did not correspond particularly well to exposure potential classifications. There are several possible reasons for this outcome. Applicators using OPs frequently are more likely to be familiar with label requirements on safe handling of the products. They may take extra precautions because they know they are at risk. Conversely, those using OPs infrequently, such as those in the low exposure potential category, may not handle the materials as carefully because they do not perceive themselves to be at risk. Also, supervisors may have assigned subjects they consider more experienced and/or more careful with pesticides in general to the

task of OP application. In fact, 3 of the subjects in the low exposure potential group had only 1 or 2 years experience working with insecticides of any type.

Effects of exposure on NES2 test performance

In the ANOVA for single sessions, wherein ChE levels and NES2 performance between exposed and control subjects were compared, statistical significance was shown in 7 comparisons, each with regard to response time, and in each case controls performed more poorly than the exposed subjects. At the .05 significance level, 6 comparisons would be expected to show significance by chance alone. The simple reaction time test, a third test of response time, did not show significant differences between controls and the exposed group, but again means were lower for the exposed group than the control group on each date. This indicates that the control group may have had a slower a priori response time than the exposed group. If there had been a real effect of exposure to OPs on reaction time, however, the exposed group should have exhibited larger differences in mean response time during sessions 2, 3, and 4, the sessions throughout which exposure occurred. However, the mean was consistent (approximately 360 milliseconds) throughout the study, with standard deviations from 33.5 to 40, indicating that OP exposure and behavioral performance were not truly related in this study.

In the repeated measures design, 5 behavioral performance scores were

shown to be statistically significant, although on different tests, none of which involved reaction time. Two to 3 comparisons out of the total of 50 would be expected to occur by comparisons would be expected to be statistically significant by chance alone at the 0.05 level. Significance was not consistent among comparisons and did not make intuitive sense. For example, the exposed groups were detected as being different from each other over time with respect to scores on number of errors on the fingertapping test, anger mean score, and fatigue mean score; however, the scores were not statistically different from those of the control groups on those tests. Neither the direction of significance nor the pattern over time were consistent with impairment of neurobehavioral function due to OP exposure in this study.

Direction of significance in the linear regressions of ChE percent change from baseline by NES2 scores was generally consistent with impairment of neurobehavioral function in the exposed group. However, at the .05 level, 8 out of the 160 comparisons would be expected to be significant by chance alone. Although 1 test (number of errors on fingertapping, session 3) was significantly different for both plasma and RBC ChE, the overall pattern of significance did not appear to provide support for an effect of OP exposure. F statistics for the same tests in different sessions during the period of OP use did not approach significance.

It is concluded that the direction and pattern of significance in all of the statistical methods used to analyze the data in this study cannot be attributed

to an effect of OP exposure. Thus this study did not demonstrate adverse behavioral effects in workers with seasonal exposure to low to moderate levels of OP exposure.

Most of the available literature on neurobehavioral effects of OPs focuses on acute exposures. Using a matched pairs approach, Savage *et al.* (1988) found persistent neurobehavioral differences in individuals with prior histories of acute OP poisoning. Although the cases in that study were chosen from hospital records of poisoning cases and thus the decrement level of ChE at the time of poisoning could not be determined, clinical symptoms requiring hospitalization are usually not associated with depletions of less than 50% of either plasma or RBC ChE (Namba, 1971), and clinical recovery occurs by the time RBC ChE returns to 30% of baseline (Kipling and Cruickshank, 1985). Tabershaw and Cooper (1986) and Rosenstock *et al.* (1991) also demonstrated impairment of neurobehavioral function in agricultural workers previously hospitalized for OP poisoning. In another group of accidental poisoning victims, Holmes and Gaon found behavioral effects associated with a decrease of 67% in RBC ChE compared to values for controls who were not exposed. Bowers *et al.* (1964) detected behavioral differences at a 60% to 90% decrease in whole blood ChE.

The current study was designed specifically to investigate effects of chronic exposure to low to moderate OP exposure. During the course of the study, 38 of the 42 exposed subjects never exhibited a drop of 25% or more.

The largest depletion of ChE detected in the present study was 46.8% plasma ChE and 39.0% RBC ChE; even these subjects would most likely not have been classified as poisoned. Results of the correlation analysis revealed poorer performance on the simple reaction time test, but better performance on a section of the fingertapping test (number of errors) concomitant with the decrease of 25% or more exhibited by 10 subjects during the course of the study. However, although 1 of the affected tests measured reaction time, neither of the other 2 NES2 tests measuring reaction time showed a significant correlation with ChE percent decrease. Still, taken together with results of the above-listed studies of neurobehavioral performance in poisoned subjects, the correlation analysis in this study may provide some insight into the ChE decrease levels at which behavioral impairment begins to be demonstrated.

A few researchers have investigated the potential effects of exposure to non-poisoning levels of OPs. Maizlish *et al.* (1987) found no adverse behavioral effects in a group of applicators compared to controls after a single application of an OP. The workers in that study had an extremely low level of exposure, as they used only granular diazinon (not a highly absorbable form of the compound); wore disposable coveralls, rubber boots, rubber gloves, and respirators while in the treatment areas as well as face shields while loading the insecticide into application equipment; and were under the direct supervision of crew chiefs. Levin *et al.* (1976) and Rodnitzky *et al.* (1975) examined farmers and structural pest control operators and found no effects associated with OP

exposure with respect to depression, memory, vigilance, signal processing time, language, or proprioception. RBC ChE mean levels in these studies did not differ between the exposed and control groups; mean plasma ChE in the exposed group was 22% lower than in the control group.

Exposure levels attained by subjects in the current study represent an intermediate level between high dose exposures studied by Savage *et al.* (1988), Tabershaw and Cooper (1986), and Rosenstock *et al.* (1991), and the extremely low exposure studied by Maizlish *et al.* (1987). This level of exposure is probably a good approximation of the exposure level of the majority of applicators using OPs over long periods of time. Therefore, the findings have implications for other workers in agriculture, structural pest control, specialized fields of pest control such as forest and mosquito treatment, and homeowners who treat fruit crops or vegetable and flower gardens.

Selection of appropriate control groups was problematic in several of the previous studies. Maizlish *et al.* (1987) used supervisors and office workers as controls. Rosenstock *et al.* (1991) solicited friends of the previously hospitalized exposed subjects as controls; they were matched for age only. The investigations reported by Levin *et al.* (1976) and Rodnitzky *et al.* (1975) used farmers matched for age and educational background as controls, but testing for some of the controls was conducted preseason as opposed to midseason for the rest of the control group and for the exposed subjects. The heterogeneity of the exposed group (some were farmers, while others were structural pest control

operators) for these 2 studies also may have confounded interpretation of results. By contrast, the present study not only used a control group matched for occupational similarity, but exposed subjects also served as their own controls since the study employed a repeated measures design.

The present study did not demonstrate adverse behavioral effects in workers with seasonal exposure to low to moderate levels of OPs. Due to the low exposure levels of the present investigation, it is not surprising that behavioral impairment detected in the high dose studies of Bowers *et al.* (1964), Tabershaw and Cooper (1986), Savage *et al.* (1988), and Rosenstock *et al.* (1991) did not occur in this study. The results of this study are consistent with the results of other studies on low-level exposure reported by Maizlish *et al.* (1987), Rodnitzky *et al.* (1975), and Levin *et al.* (1976), and partially in concurrence with the results of Korsak and Sato (1977). Comparability between the neurobehavioral tests used in this and other studies is, of course, uncertain.

Test methodology

While several studies currently underway are employing the NES2 (or other versions) as the measure of neurobehavioral performance, most earlier researchers chose other tests whose comparability to NES is not precisely understood. The movement toward use of the NES2 as the tool of choice should alleviate this problem in the future.

Sources of bias

A number of potentially confounding relationships involving age, education, exercise, use of insecticides at home, and use of alcohol and drugs were identified. The somewhat older age structure of the control group may have been a confounding factor in this study, but the other factors were quite similar between the exposed and control groups and can therefore probably be discounted.

Specific OPs used as well as the schedule of use of OPs in this study varied from applicator to applicator. The decision to treat with a particular insecticide depends on the species, population level and life stage of the pest, particular site conditions, availability of products and of application equipment, economic considerations, company policy, and, in some cases, customer demand. These are real world conditions of use but they introduce more variability into the study with regard to exposure.

Participants were asked to use the same pesticide application practices they had employed in the past. However, in accordance with IRB requirements, all participants had been informed of the purpose of the study and thus were aware that OPs had been shown in past studies to be associated with neuro-behavioral effects. This knowledge may have caused exposed subjects to exercise more care than usual when working with insecticides in general (most participants did not know which of the materials they used were OPs), and so

may have lowered the exposure potential.

Both exposed and control participants generally had a high level of interest in the project. All were no doubt motivated to some extent by the prospect of receiving a \$100 honorarium for completing all sessions, but most expressed a genuine interest in the results throughout the test period. Such motivation is generally believed to be desirable for studies on behavior; the CDC/NIOSH panel stressed the importance of having subjects give their best effort on the NES2 tests.

Subject selection should be considered with regard to representativeness of the general population of OP applicators. Firms allowing their employees to participate in the study generally had good reputations for safety with regard to worker training and application practices. Applicators also tended to have a higher level of education than may be representative of the industry as a whole. Several of these firms rely on their reputation for environmental safety and the advanced degrees of their managers and staff to attract customers. Those firms with poor safety records, less highly trained and/or educated workers, and less rigorous work standards generally did not want their employees to participate. It is likely that both the relatively high level of safety training and the widespread use of appropriate personal protective gear by participants provided extra protection to these workers compared to workers of nonparticipating firms, thereby maintaining their real OP dose at very low levels. Thus the pest control practices of participants could be expected to be

at the upper end of the spectrum compared to those of the industry at large.

While 2 of the participating firms required ChE testing of all OP applicators as a condition of employment, the other 16 firms had no such monitoring program. Several of the firms were owner-operated; thus these applicators were deriving the benefit of regular ChE testing for themselves as well as their employees, which they were not willing to pay for on their own. Having blood drawn on a regular basis may serve as a reminder to applicators of the hazards they face and may lead to their taking extra precautions when handling pesticides in general. In at least one of the participating firms, ChE depression below 75% of baseline is considered a very serious matter; the second instance of such a depression is grounds for dismissal.

In general, the education and training, use of protective gear, and high level of motivation to avoid overexposure in the participating subjects may set them apart from nonparticipating applicators. Although subjects were asked to continue to use the same work practices as in the past, it is possible some subjects were more cautious when using pesticides during this study.

Conclusions

This is the first study to investigate OP effects in workers a.) exposed to relatively low doses, and b.) tested over the entire spray season rather than at one point in time. The results indicate that seasonal exposure to low to moderate doses does not appear to have a cumulative effect on behavior. Thus, at the exposure levels experienced by subjects in this study, OPs do not appear to alter the behavioral functions measured, *i.e.*, perception, motor reflexes, memory, and affect. Based on the sensitivity of the methods in this study, if behavioral changes do occur at this level of exposure they are very small: less than a 4 to 15% difference in memory and perceptual-motor functions, and less than a 15 to 23% difference in affect.

Because the study population was probably more careful than the industry at large, the possibility of neurobehavioral effects in workers with less rigorous safety practices, and thus with higher exposure to OPs, should not be disregarded. Previous research has shown clearly that high acute exposures can cause detrimental neurobehavioral changes in insecticide applicators. The use of appropriate personal protective gear and the high levels of education and safety training of the exposed group in the current study undoubtedly contributed to their low levels of exposure and to the lack of effects detected.

Future research needs

This study investigated the neurobehavioral effects in well-trained, safety-conscious applicators exposed to relatively low levels of OPs. Around the world, workers apply more concentrated formulations of OPs more frequently, with less protective clothing and little safety training. An investigation of the possible effects of a higher level of OP exposure over time is needed. Additionally, further investigation into the level at which neurobehavioral impairment begins upon an acute decrease in ChE level is called for. A group of subjects applying OPs on a more regular schedule would minimize study design problems related to differences in frequency of exposure. Because of the subtlety of effects, a large population size should be chosen to optimize the probability of detecting effects.

Appendix A.--Employer questionnaire

NAME OF COMPANY

ADDRESS

CONTACT (SUPERVISOR)

1. When do you begin applying organophosphates?
2. How many will participate?
3. When are you at full staff?
4. What is your heaviest use time period?
5. Convenient facility near you?

Appendix B.--Personal data form.

Code number: _____

PARTICIPANT DATA SHEET

Full name: _____

Company: _____

Home address: _____

Social security number: _____

Home telephone number: _____

Highest level of education completed: _____

Appendix C.--Medical history form.

CONFIDENTIAL MEDICAL HISTORY FORM

Please print your name (first, middle, last) and telephone number:

Certain medical conditions are known to affect some of the parameters being measured in this study. Your answers to the following questions will be used in final analysis of the data collected, as well as in helping to determine whether your participation in the project would be appropriate. Answering "yes" to any of the questions will not necessarily disqualify you from participating. All of the information collected on this form will be kept strictly confidential and will be destroyed as soon as the study is completed.

Have you ever been told by a medical doctor that you have any of the following conditions?

condition	yes	no
organophosphate poisoning	—	—
central nervous system pathology	—	—
peripheral neuropathy	—	—
any muscular disease	—	—
permanent muscle injury	—	—
seizure disorder	—	—
arthritis	—	—
liver disease	—	—
head injury	—	—

Appendix D.--Consent form

Project Title: Chronic organophosphate exposure: Neurobehavioral effects.

I state that I am over 18 years of age, in good physical health, and wish to participate in a program of research being conducted by Amy Brown at the Graduate School, University of Maryland, College Park, Marine-Estuarine-Environmental Studies Program.

The purpose of the research is to investigate the effects of low to moderate exposure to organophosphate insecticides on neurobehavioral parameters. The procedures involve a blood sample being drawn by a laboratory technician on 5 separate occasions, and the performance of 5 computerized neurobehavioral test sessions (approximately 40 minutes each) plus 1 practice session at the beginning of the study. A brief questionnaire will be administered during the last test session. The time course of this study will be from the Spring of 1990 through winter of 1990-1991.

If I should exhibit a 25% drop in ChE at any time, the safety officer of my firm will be informed and may notify me, for my own safety, to refrain from applying organophosphates until the level returns to within the normal range. These are the only individual results that may be divulged to my employer. All individual results of neurobehavioral tests, as well as my answers to the post-season questionnaire, will be kept strictly confidential.

I understand that some people experience transient discomfort or hematoma (bruising) from having their blood drawn. I understand that sterile techniques will be used during all procedures associated with blood sampling, and that AIDS, hepatitis, and other blood-borne diseases cannot be transmitted by such sampling.

I understand that my risk of exposure to organophosphates will not be increased, and may actually be decreased, through participation in this study. I understand that the researcher hopes to learn more about effects of organophosphates on the nervous system through the combined results of this investigation. I understand that I am free to ask questions or to withdraw from participation at any time without penalty.

I understand that the University of Maryland does not provide any medical or hospitalization insurance coverage for participants in the research study, nor will the University of Maryland provide compensation for any injury sustained as a result of participation in this research study except as required by law.

Name, address, and phone number of Principal Investigator:

Amy E. Brown
Department of Entomology
University of Maryland
College Park, MD 20742
301-454-7127

Signature

Date

Appendix E.--Questionnaire for exposed subjects

 E

DO NOT WRITE YOUR NAME ON THIS QUESTIONNAIRE! You have been assigned a code number in order to maintain confidentiality.

Answers to the following questions will be used in the final analysis of the study to help draw appropriate conclusions from the results of the tests you have taken throughout the season. All of the information collected will be kept strictly confidential. In the final report, answers to the questions will be pooled so that no single employee can be identified.

1. How long have you been employed in your current job?
2. Before this job, did your work involve insecticides? Yes No

If you answered yes, please tell me how many years you have worked with insecticides:

-
3. Are you:
 a registered technician
 a certified applicator
 neither of the above
4. When you first began working with insecticides, how much training on insecticide safety had you received?
 More than 1 full day of safety training.
 1 day of safety training.
 Some safety training, but less than 1 day.
 No safety training.
5. During the entire time you have been working with insecticides, how much training on insecticide safety have you received?
 More than 1 day of safety training per year.
 1 day of safety training per year.
 Some safety training, but less than 1 day per year.
 No safety training.
- E**
6. During the past year, how often did you use insecticides at home?
 Once a week.
 Every other week.
 Once a month.
 Never.

Appendix E. (cont'd.)

- _____ E
7. During the past year, did your job include mixing insecticides from concentrates?
____ Yes ____ No

If you answered Yes to question number 7, which of the following did you usually wear when mixing insecticides? (If you answered No, skip to question #8.)

- ____ Disposable outfit with long pants and long sleeves.
- ____ Company outfit with long pants and long sleeves.
- ____ Company outfit with long pants and short sleeves.
- ____ Your own clothing -- long pants and a long-sleeved shirt.
- ____ Your own clothing -- long pants and a short-sleeved shirt.
- ____ Your own clothing -- shorts and a short-sleeved shirt.
- ____ Gloves -- cotton.
- ____ Gloves -- rubber or neoprene.
- ____ No gloves.
- ____ A hat.
- ____ Goggles.
- ____ A respirator or gas mask.
- ____ Rubber or neoprene boots.
- ____ Leather work boots, leather sneakers, or leather shoes.
- ____ Sandals.
- ____ Canvas shoes (tennis shoes, sneakers).

Appendix E. (cont'd.)

E

8. Which of the following did you usually wear when applying insecticides?

- ☐ Disposable outfit with long pants and long sleeves.
- ☐ Company outfit with long pants and long sleeves.
- ☐ Company outfit with long pants and short sleeves.
- ☐ Your own clothing -- long pants and a long-sleeved shirt.
- ☐ Your own clothing -- long pants and a short-sleeved shirt.
- ☐ Your own clothing -- shorts and a short-sleeved shirt.
- ☐ Gloves -- cotton.
- ☐ Gloves -- rubber or neoprene.
- ☐ No gloves.
- ☐ A hat.
- ☐ Goggles.
- ☐ A respirator or gas mask.
- ☐ Rubber or neoprene boots.
- ☐ Leather work boots, leather sneakers, or leather shoes.
- ☐ Sandals.
- ☐ Canvas shoes (tennis shoes, sneakers).

9. How would you classify the amount of exercise you got during the past year?

- ☐ I did not exercise other than performing my job.
- ☐ I exercised moderately in addition to my job (weekly walking, bicycling, etc.)
- ☐ I got a lot of exercise (daily jogging, swimming, aerobics, etc.)

Appendix E. (cont'd.)

10. During the past year, how often did you use each of the following substances? E

Caffeinated beverages (coffee, tea, cola drinks, etc.)

☐ Every day ☐ Once or twice a week ☐ Once or twice a month ☐ Never

Tobacco (cigarettes, cigars, pipe, chewing tobacco, etc.)

☐ Every day ☐ Once or twice a week ☐ Once or twice a month ☐ Never

Alcohol (beer, wine, hard liquor)

☐ Every day ☐ Once or twice a week ☐ Once or twice a month ☐ Never

11. During the past year, how often did you use each of the following substances?

Cocaine

☐ Every day ☐ Once or twice a week ☐ Once or twice a month ☐ Never

Marijuana

☐ Every day ☐ Once or twice a week ☐ Once or twice a month ☐ Never

Other "recreational" drugs -- please list and indicate how often you used each during the past year:

Drug: _____

☐ Every day ☐ Once or twice a week ☐ Once or twice a month

Drug: _____

☐ Every day ☐ Once or twice a week ☐ Once or twice a month

Drug: _____

☐ Every day ☐ Once or twice a week ☐ Once or twice a month

Appendix E. (cont'd.)

12. Please list any prescription medications you took regularly during the past year and indicate how often you used each: E

Drug: _____

☐ Every day ☐ Once or twice a week ☐ Once or twice a month

Drug: _____

☐ Every day ☐ Once or twice a week ☐ Once or twice a month

Drug: _____

☐ Every day ☐ Once or twice a week ☐ Once or twice a month

Drug: _____

☐ Every day ☐ Once or twice a week ☐ Once or twice a month

Drug: _____

☐ Every day ☐ Once or twice a week ☐ Once or twice a month

Drug: _____

☐ Every day ☐ Once or twice a week ☐ Once or twice a month

Appendix E. (cont'd.)

 E

13. If you used any pesticides as part of your job this year or in the past, please check them off on the list below.

PESTICIDE	USED IN PAST YEAR	USED IN PREVIOUS YEARS (list years)	PESTICIDE	USED IN PAST YEAR	USED IN PREVIOUS YEARS (list years)
Acti-Dione			BT (Dipel)		
Banvel			Bayleton		
Bidrin			Carbamate		
Chipco 26019			Cygon		
Daconil			Dacthal		
diazinon			dicamba		
Dimilin			Dursban		
Dyrene			Garlon		
Insecticidal soap			Kelthane		
Lindane			LSR		
malathion			Marlate		
Mavrik			methoxychlor		
Morestan			Oftanol		
Orthene			Pydrin		
Pentac			PGF		
pyrethrins			rotenone		
Roundup			Rubigan		
Sevin			Subdue		
Tempo II			Terasan 1991		
Turcam			Vendex		
Other:			Other:		
Other:			Other:		
Other:			Other:		

Appendix F.--Questionnaire for control subjects

DO NOT WRITE YOUR NAME ON THIS QUESTIONNAIRE! You have been assigned a code number in order to maintain confidentiality.

Answers to the following questions will be used in the final analysis of the study to help draw appropriate conclusions from the results of the tests you have taken throughout the season. All of the information collected will be kept strictly confidential. In the final report, answers to the questions will be pooled so that no single employee can be identified.

1. How long have you been employed in your current job? _____
2. Before this job, did your work involve insecticides? _____ Yes _____ No

If you answered yes, please tell me how many years you worked with insecticides:

- _____
3. During the past year, how often did you use insecticides at home?

___ Once a week.
___ Every other week.
___ Once a month.
___ Never.

4. How would you classify the amount of exercise you got during the past year?

___ I did not exercise other than performing my job.
___ I exercised moderately in addition to my job (weekly walking, bicycling, etc.)
___ I got a lot of exercise (daily jogging, swimming, aerobics, etc.)

Appendix F. (cont'd.)

5. During the past year, how often did you use each of the following substances? _____C

Caffeinated beverages (coffee, tea, cola drinks, etc.)

___ Every day ___ Once or twice a week ___ Once or twice a month ___ Never

Tobacco (cigarettes, cigars, pipe, chewing tobacco, etc.)

___ Every day ___ Once or twice a week ___ Once or twice a month ___ Never

Alcohol (beer, wine, hard liquor)

___ Every day ___ Once or twice a week ___ Once or twice a month ___ Never

6. During the past year, how often did you use each of the following substances?

Cocaine

___ Every day ___ Once or twice a week ___ Once or twice a month ___ Never

Marijuana

___ Every day ___ Once or twice a week ___ Once or twice a month ___ Never

Other "recreational" drugs -- please list and indicate how often you used each during the past year:

Drug: _____

___ Every day ___ Once or twice a week ___ Once or twice a month

Drug: _____

___ Every day ___ Once or twice a week ___ Once or twice a month

Drug: _____

___ Every day ___ Once or twice a week ___ Once or twice a month

Appendix F. (cont'd.)

7. Please list any prescription medications you took regularly during the past year and indicate how often you used each: C

Drug: _____

☐ Every day ☐ Once or twice a week ☐ Once or twice a month

Drug: _____

☐ Every day ☐ Once or twice a week ☐ Once or twice a month

Drug: _____

☐ Every day ☐ Once or twice a week ☐ Once or twice a month

Drug: _____

☐ Every day ☐ Once or twice a week ☐ Once or twice a month

Drug: _____

☐ Every day ☐ Once or twice a week ☐ Once or twice a month

Drug: _____

☐ Every day ☐ Once or twice a week ☐ Once or twice a month

Appendix F. (cont'd.)

E

13. If you used any pesticides as part of your job this year or in the past, please check them off on the list below.

PESTICIDE	USED IN PAST YEAR	USED IN PREVIOUS YEARS (list years)	PESTICIDE	USED IN PAST YEAR	USED IN PREVIOUS YEARS (list years)
Acti-Dione			BT (Dipel)		
Banvel			Bayleton		
Bidrin			Carbamate		
Chipco 26019			Cygon		
Daconil			Dacthal		
diazinon			dicamba		
Dimilin			Dursban		
Dyrene			Garlon		
Insecticidal soap			Kelthane		
Lindane			LSR		
malathion			Marlate		
Mavrik			methoxychlor		
Morestan			Oftanol		
Orthene			Pydrin		
Pentac			PGF		
pyrethrins			rotenone		
Roundup			Rubigan		
Sevin			Subdue		
Tempo II			Terasan 1991		
Turcam			Vendex		
Other:			Other:		
Other:			Other:		
Other:			Other:		

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